Complex Analysis II

Lecture notes following Stein–Shakarchi, Ch. 8–10 and Appendices A–B

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Chapter 8. Conformal Mappings

Overview

This chapter is geometric and "global": we study mapping properties of holomorphic functions, motivated by the question

> *Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?*

(The inverse will automatically be holomorphic.) The most fruitful choice is $V = \mathbb{D}$: *what conditions on Ω guarantee a holomorphic bijection $\Omega \to \mathbb{D}$?* (For $V = \mathbb{C}$ only $U = \mathbb{C}$ is possible.) We first give explicit examples (the upper half-plane maps to the disc by a fractional linear transformation; compositions of elementary maps) and apply them to the Dirichlet problem. Then the Schwarz lemma determines the automorphisms of the disc and of the upper half-plane. The heart of the chapter is the *Riemann mapping theorem*: Ω is conformally equivalent to $\mathbb{D}$ whenever it is simply connected and not all of $\mathbb{C}$, with no assumption on $\partial\Omega$. For polygons the map is given by the *Schwarz–Christoffel formula*; for rectangles it involves elliptic integrals, which lead to the doubly periodic functions of Chapter 9.

1 Conformal equivalence and examples

Motivation. When are two regions "the same" for complex analysis? When a holomorphic bijection maps one onto the other. Such maps transport problems, for example the Dirichlet problem, from a complicated region to a simple one such as the disc or the half-plane.

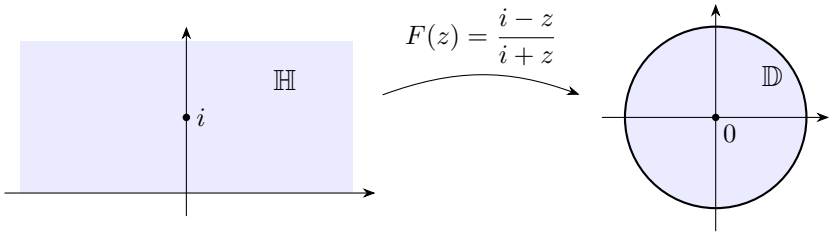


Figure. *The Cayley-type map F sends the upper half-plane $\mathbb{H}$ onto the unit disc $\mathbb{D}$, the real axis onto the circle minus -1, and i to 0; problems on $\mathbb{H}$ and on $\mathbb{D}$ are therefore equivalent.*

A bijective holomorphic $f : U \to V$ is a **conformal map** or **biholomorphism**; then U and V are **conformally equivalent** (**biholomorphic**).

Proposition 1.1

If $f : U \to V$ is holomorphic and injective, then $f'(z) \neq 0$ for all $z \in U$. In particular, the inverse of f defined on its range is holomorphic, and thus the inverse of a conformal map is also holomorphic.

Proof. Suppose $f'(z_0) = 0$. Then $f(z) - f(z_0) = a(z - z_0)^k + G(z)$ near z_0, with $a \neq 0$, $k \geq 2$, and G vanishing to order $k+1$ at z_0. For small w write $f(z) - f(z_0) - w = F(z) + G(z)$ with $F(z) = a(z - z_0)^k - w$. On a small circle about z_0, $|G| < |F|$, and F has at least two zeros inside, so by Rouché's theorem $f(z) - f(z_0) - w$ has at least two zeros there. Since $f'(z) \neq 0$ for $z \neq z_0$ near z_0 (zeros of f' are isolated), these roots are distinct, so f is not injective.

Let $g = f^{-1}$ on the range V, $w_0 = f(z_0)$, $w = f(z)$ close to w_0, $w \neq w_0$. Then

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w-w_0}{g(w)-g(w_0)}} = \frac{1}{\frac{f(z)-f(z_0)}{z-z_0}}.$$

Since $f'(z_0) \neq 0$, letting $z \to z_0$ shows g is holomorphic at w_0 with $g'(w_0) = 1/f'(g(w_0))$. ■

So U, V are conformally equivalent iff there are holomorphic $f : U \to V$, $g : V \to U$ with $g(f(z)) = z$ and $f(g(w)) = w$. (Some authors call f conformal when merely $f' \neq 0$, e.g. z^2 on $\mathbb{C} - \{0\}$, which is not injective. The condition $f'(z) \neq 0$ means f is a local bijection, and such a map preserves angles between curves.)

1.1 The disc and upper half-plane

The **upper half-plane** is $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$. Let

$$F(z) = \frac{i - z}{i + z} \qquad \text{and} \qquad G(w) = i\frac{1 - w}{1 + w}.$$

Theorem 1.2

The map $F : \mathbb{H} \to \mathbb{D}$ is a conformal map with inverse $G : \mathbb{D} \to \mathbb{H}$.

Proof. Both maps are holomorphic in their domains. Points of $\mathbb{H}$ are closer to i than to $-i$, so $|F(z)| < 1$. For $w = u + iv \in \mathbb{D}$,

$$\mathrm{Im}(G(w)) = \mathrm{Re}\left(\frac{1 - u - iv}{1 + u + iv}\right) = \mathrm{Re}\,\frac{(1 - u - iv)(1 + u - iv)}{(1 + u)^2 + v^2} = \frac{1 - u^2 - v^2}{(1 + u)^2 + v^2} > 0.$$

Finally

$$F(G(w)) = \frac{i - i\frac{1-w}{1+w}}{i + i\frac{1-w}{1+w}} = \frac{1 + w - 1 + w}{1 + w + 1 - w} = w,$$

and similarly $G(F(z)) = z$. ■

F is holomorphic on $\mathbb{C}$ except at $-i$, hence continuous on $\mathbb{R} = \partial\mathbb{H}$, and $|F(x)| = 1$ for real x. With $x = \tan t$, $t \in (-\pi/2, \pi/2)$,

$$F(x) = \frac{1 - x^2}{1 + x^2} + i\frac{2x}{1 + x^2} = \cos 2t + i \sin 2t = e^{2it},$$

so F maps $\mathbb{R}$ onto the circle minus -1, traversed from -1 first through the lower half; -1 corresponds to the point at infinity of $\mathbb{H}$.

Remark

Maps $z \mapsto \frac{az+b}{cz+d}$ ($a, b, c, d \in \mathbb{C}$, denominator not a multiple of the numerator) are **fractional linear transformations**; they reappear as the automorphisms of the disc and of $\mathbb{H}$ (Theorems 2.2, 2.4).

1.2 Further examples

Example 1

Translations and dilations. For $h \in \mathbb{C}$, $z \mapsto z+h$ is a conformal map of $\mathbb{C}$ (inverse $w \mapsto w-h$); for real h it is also a conformal map of $\mathbb{H}$. For $c \neq 0$, $z \mapsto cz$ is conformal on $\mathbb{C}$ (inverse $w \mapsto c^{-1}w$): a rotation if $c = e^{i\varphi}$, a dilation if $c > 0$, a dilation by $|c|$ followed by rotation by π if $c < 0$.

Example 2

For a positive integer n, $z \mapsto z^n$ is conformal from the sector $\{0 < \arg z < \pi/n\}$ to $\mathbb{H}$, with inverse $w \mapsto w^{1/n}$ (principal branch). More generally, for $0 < \alpha < 2$, $f(z) = z^\alpha$ maps $\mathbb{H}$ to the sector $S = \{0 < \arg w < \alpha\pi\}$: with the branch obtained by deleting the positive real axis and $z = re^{i\theta}$, $0 < \theta < \pi$, $f(z) = |z|^\alpha e^{i\alpha\theta}$; the inverse is $w^{1/\alpha}$ with $0 < \arg w < \alpha\pi$. Composing with Example 1, $\mathbb{H}$ maps conformally onto any infinite sector. On the boundary: as x goes from $-\infty$ to 0, $f(x)$ goes from $\infty e^{i\alpha\pi}$ to 0 along $\arg z = \alpha\pi$; as x goes from 0 to ∞, $f(x)$ goes from 0 to ∞ along $\mathbb{R}$.

Example 3

$f(z) = \frac{1+z}{1-z}$ maps the upper half-disc $\{|z| < 1,\ y > 0\}$ conformally onto the first quadrant $\{u > 0, v > 0\}$: for $z = x + iy$,

$$f(z) = \frac{1-(x^2+y^2)}{(1-x)^2+y^2} + i\frac{2y}{(1-x)^2+y^2}.$$

The inverse $g(w) = \frac{w-1}{w+1}$ is holomorphic in the first quadrant, maps it into $\mathbb{D}$ (the distance from w to -1 exceeds that to 1) with $\operatorname{Im} g(w) > 0$. On the boundary, for $z = e^{i\theta}$, $f(z) = \frac{e^{-i\theta/2}+e^{i\theta/2}}{e^{-i\theta/2}-e^{i\theta/2}} = \frac{i}{\tan(\theta/2)}$, so as θ goes from 0 to π, $f(e^{i\theta})$ travels down the imaginary axis from ∞ to 0; f maps $(-1, 1)$ increasingly onto $(0, \infty)$, $f(0) = 1$.

Example 4

$z \mapsto \log z$, the branch obtained by deleting the negative imaginary axis, maps $\mathbb{H}$ onto the strip $\{w = u + iv : 0 < v < \pi\}$: for $z = re^{i\theta}$, $-\pi/2 < \theta < 3\pi/2$, $\log z = \log r + i\theta$. The inverse is $w \mapsto e^w$. As x goes from $-\infty$ to 0, $f(x)$ goes from $\infty + i\pi$ to $-\infty + i\pi$; as x goes from 0 to ∞, $f(x)$ goes from $-\infty$ to ∞ along $\mathbb{R}$.

Example 5

$z \mapsto \log z$ also maps the half-disc $\{|z| < 1, y > 0\}$ onto the half-strip $\{u < 0,\ 0 < v < \pi\}$: $(0, 1)$ goes to $(-\infty, 0)$, the upper half-circle from 1 to -1 to the segment from 0 to $i\pi$, and $(-1, 0)$ to the half-line from $i\pi$ to $-\infty + i\pi$.

Example 6

$f(z) = e^{iz}$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ onto the half-disc $\{|w| < 1,\ u > 0\}$, since $e^{iz} = e^{-y}e^{ix}$. The boundary goes: from $\pi/2 + i\infty$ to $\pi/2$, f goes from 0 to i; from $\pi/2$ to $-\pi/2$, f goes from i to $-i$ along the half-circle; from $-\pi/2$ to $-\pi/2 + i\infty$, from $-i$ back to 0. (This is closely related to the inverse of Example 5.)

Example 7

$f(z) = -\frac{1}{2}(z+1/z)$ maps the half-disc $\{|z| < 1, y > 0\}$ conformally onto $\mathbb{H}$. On the boundary: as x goes from 0 to 1, $f(x)$ goes from $-\infty$ to -1; on the half-circle $z = e^{i\theta}$, $f(z) = -\cos\theta$ runs from -1 to 1 as θ goes from 0 to π; and as x goes from -1 to 0, $f(x)$ goes from 1 to ∞.

Example 8

$f(z) = \sin z$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ conformally onto $\mathbb{H}$. With $\zeta = e^{iz}$,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = -\frac{1}{2}\Big(i\zeta + \frac{1}{i\zeta}\Big),$$

so f is the map of Example 6, followed by multiplication by i (rotation by $\pi/2$, giving the upper half-disc), followed by Example 7. As z goes from $-\pi/2 + i\infty$ to $-\pi/2$, f goes from $-\infty$ to -1; on $[-\pi/2, \pi/2]$, f runs over $[-1, 1]$; from $\pi/2$ to $\pi/2 + i\infty$, f goes from 1 to ∞.

1.3 The Dirichlet problem in a strip

The **Dirichlet problem** in Ω is

$$\triangle u = 0 \text{ in } \Omega, \qquad u = f \text{ on } \partial\Omega, \tag{1}$$

$\triangle = \partial^2/\partial x^2 + \partial^2/\partial y^2$, f given on $\partial\Omega$. For the disc and $\mathbb{H}$ it is solved by convolution with Poisson kernels. We solve it for a strip using conformal maps and the solution in the disc.

Lemma 1.3

Let V and U be open sets in $\mathbb{C}$ and $F : V \to U$ a holomorphic function. If $u : U \to \mathbb{C}$ is a harmonic function, then $u \circ F$ is harmonic on V.

Proof. The question is local, so take U a disc. There is G holomorphic on U with $\operatorname{Re} G = u$ (every harmonic function in a disc is the real part of a holomorphic function). Then $u \circ F = \operatorname{Re}(G \circ F)$ with $G \circ F$ holomorphic, so $u \circ F$ is harmonic. ∎

Let $\Omega = \{x + iy : x \in \mathbb{R},\ 0 < y < 1\}$, with boundary data $u(x, 0) = f_0(x)$, $u(x, 1) = f_1(x)$, f_0, f_1 continuous and vanishing at infinity. Transfer the problem to $\mathbb{D}$ by

$$F(w) = \frac{1}{\pi}\log\Big(i\frac{1-w}{1+w}\Big) : \mathbb{D} \to \Omega, \qquad G(z) = \frac{i - e^{\pi z}}{i + e^{\pi z}} : \Omega \to \mathbb{D},$$

compositions of maps above, conformal and mutually inverse. F maps the lower half-circle onto $i + \mathbb{R}$ and the upper half-circle onto $\mathbb{R}$: as φ goes from $-\pi$ to 0, $F(e^{i\varphi})$ goes from $i + \infty$ to $i - \infty$, and as φ goes from 0 to π, $F(e^{i\varphi})$ goes from $-\infty$ to ∞. Define $\tilde{f}_1(\varphi) = f_1(F(e^{i\varphi}) - i)$ for $-\pi < \varphi < 0$ and $\tilde{f}_0(\varphi) = f_0(F(e^{i\varphi}))$ for $0 < \varphi < \pi$; since f_0, f_1 vanish at infinity, $\tilde{f}$ (equal to $\tilde{f}_1$, $\tilde{f}_0$ on the two half-circles and 0 at $\varphi = \pm\pi, 0$) is continuous on the circle. The solution in the disc is the Poisson integral, for $w = re^{i\theta}$,

$$\tilde{u}(w) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi)\tilde{f}(\varphi)\,d\varphi = \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta - \varphi)\tilde{f}_1(\varphi)\,d\varphi + \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta - \varphi)\tilde{f}_0(\varphi)\,d\varphi,$$

$P_r(\gamma) = \frac{1-r^2}{1-2r\cos\gamma+r^2}$ the Poisson kernel. By Lemma 1.3, $u(z) = \tilde{u}(G(z))$ is harmonic in the strip, with the correct boundary values. For $z = iy$, $0 < y < 1$, with $re^{i\theta} = G(iy)$, a change of variables gives

$$\frac{1}{2\pi}\int_0^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)\,d\varphi = \frac{\sin\pi y}{2}\int_{-\infty}^{\infty}\frac{f_0(t)}{\cosh\pi t - \cos\pi y}\,dt, \qquad \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde{f}_1(\varphi)\,d\varphi = \frac{\sin\pi y}{2}\int_{-\infty}^{\infty}\frac{f_1(t}{\cosh\pi t +}$$

A solution vanishing at infinity is unique, so translating the boundary data by x translates the solution, and

$$u(x,y) = \frac{\sin \pi y}{2}\Big(\int_{-\infty}^{\infty} \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y}\, dt + \int_{-\infty}^{\infty} \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y}\, dt \Big),$$

convolutions with f_0 and f_1. At $y = 1/2$ the kernel is $1/\cosh \pi t$, its own Fourier transform (Example 3 in Section 2 of Chapter 3).

Remarks about the Dirichlet problem. If $F : \mathbb{D} \to \Omega$ is conformal and extends to a continuous bijection $\partial\mathbb{D} \to \partial\Omega$, then for f on $\partial\Omega$, $\tilde{f} = f \circ F$ is defined on the circle, the Poisson integral $\tilde{u}(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta - \varphi)\tilde{f}(e^{i\varphi})\, d\varphi$ solves the problem in the disc, and $u = \tilde{u} \circ F^{-1}$ is expected to solve (1). This requires:

(i) existence of a conformal map $\Phi = F^{-1} : \Omega \to \mathbb{D}$, settled in general by the Riemann mapping theorem;

(ii) its extension to a continuous bijection of the boundaries, which needs some regularity of $\partial\Omega$ (treated for polygons in Section 4). Riemann reversed the implication: to find Φ with $\Phi(z_0) = 0$, write $\Phi(z) = (z - z_0)e^{H(z)}$; $|\Phi| = 1$ on $\partial\Omega$ means $u = \operatorname{Re} H$ solves the Dirichlet problem with $u(z) = \log(1/|z - z_0|)$ on $\partial\Omega$ (u is the Green's function with source z_0). This approach requires showing Φ is a bijection, some boundary regularity, and solving the Dirichlet problem (Riemann proposed the "Dirichlet principle", whose implementation has difficulties). We prove the existence in general by different methods in Section 3.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

Motivation. To understand conformal maps of the disc we first find all of its automorphisms. The Schwarz lemma, a simple consequence of the maximum principle, forces them to be Möbius transformations, and the same holds for the upper half-plane.

A **rotation** is a map $z \mapsto cz$ with $|c| = 1$, $c = e^{i\varphi}$, φ the angle of rotation (defined modulo 2π).

Lemma 2.1 (Schwarz lemma)

Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then (i) $|f(z)| \le |z|$ for all $z \in \mathbb{D}$.
(ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation.
(iii) $|f'(0)| \le 1$, and if equality holds, then f is a rotation.

Proof. Expand $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots$ in $\mathbb{D}$; $f(0) = 0$ gives $a_0 = 0$, so $f(z)/z$ has a removable singularity at 0 and is holomorphic in $\mathbb{D}$. If $|z| = r < 1$, since $|f(z)| \le 1$, $|f(z)/z| \le 1/r$, and by the maximum modulus principle this holds for $|z| \le r$. Letting $r \to 1$ gives (i). For (ii), $f(z)/z$ attains its maximum in the interior of $\mathbb{D}$, so it is a constant c, and $|c| = 1$ (evaluate at z_0). For (iii), $g(z) = f(z)/z$ satisfies $|g| \le 1$ on $\mathbb{D}$ and $g(0) = \lim_{z\to 0} \frac{f(z)-f(0)}{z} = f'(0)$; if $|f'(0)| = 1$, then $|g(0)| = 1$, so g is constant by the maximum principle and $f(z) = cz$, $|c| = 1$. ∎

2.1 Automorphisms of the disc

A conformal map of Ω to itself is an **automorphism** of Ω; they form a group $\operatorname{Aut}(\Omega)$ under composition, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. Rotations $r_\theta : z \mapsto e^{i\theta}z$ are automorphisms of $\mathbb{D}$ (inverse $r_{-\theta}$). More interesting are

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}, \qquad \alpha \in \mathbb{C},\ |\alpha| < 1.$$

ψ_α is holomorphic in $\mathbb{D}$. If $z = e^{i\theta}$, $\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \overline{\alpha})} = -e^{-i\theta}\frac{w}{\overline{w}}$ with $w = \alpha - e^{i\theta}$, so $|\psi_\alpha| = 1$ on the circle and $|\psi_\alpha| < 1$ in $\mathbb{D}$ by the maximum modulus principle. Moreover

$$(\psi_\alpha \circ \psi_\alpha)(z) = \frac{\alpha - \frac{\alpha - z}{1 - \overline{\alpha}z}}{1 - \overline{\alpha}\frac{\alpha - z}{1 - \overline{\alpha}z}} = \frac{\alpha - |\alpha|^2 z - \alpha + z}{1 - \overline{\alpha}z - |\alpha|^2 + \overline{\alpha}z} = \frac{(1 - |\alpha|^2)z}{1 - |\alpha|^2} = z,$$

so ψ_α is its own inverse; and it interchanges 0 and α: $\psi_\alpha(0) = \alpha$, $\psi_\alpha(\alpha) = 0$.

Theorem 2.2

If f is an automorphism of the disc, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that

$$f(z) = e^{i\theta}\frac{\alpha - z}{1 - \overline{\alpha}z}.$$

Proof. There is a unique $\alpha \in \mathbb{D}$ with $f(\alpha) = 0$. Let $g = f \circ \psi_\alpha$; then $g(0) = 0$ and by the Schwarz lemma

$$|g(z)| \le |z| \qquad \text{for all } z \in \mathbb{D}. \tag{2}$$

Also $g^{-1}(0) = 0$, so $|g^{-1}(w)| \le |w|$; with $w = g(z)$,

$$|z| \le |g(z)| \qquad \text{for all } z \in \mathbb{D}. \tag{3}$$

By (2), (3), $|g(z)| = |z|$, so $g(z) = e^{i\theta}z$ by the Schwarz lemma. Replacing z by $\psi_\alpha(z)$ and using $\psi_\alpha \circ \psi_\alpha = \mathrm{id}$, $f(z) = e^{i\theta}\psi_\alpha(z)$. ∎

Corollary 2.3

The only automorphisms of the unit disc that fix the origin are the rotations.

The group $\mathrm{Aut}(\mathbb{D})$ acts *transitively*: given $\alpha, \beta \in \mathbb{D}$, $\psi_\beta \circ \psi_\alpha$ maps α to β. $\mathrm{Aut}(\mathbb{D})$ is "almost" isomorphic to the matrix group $SU(1,1)$ of 2×2 complex matrices preserving the hermitian form $\langle Z, W\rangle = z_1\overline{w_1} - z_2\overline{w_2}$ on $\mathbb{C}^2 \times \mathbb{C}^2$.

2.2 Automorphisms of the upper half-plane

With $F : \mathbb{H} \to \mathbb{D}$ from Section 1, define $\Gamma : \mathrm{Aut}(\mathbb{D}) \to \mathrm{Aut}(\mathbb{H})$ by "conjugation by F": $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$. Γ is a bijection with inverse $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$, and it preserves composition:

$$\Gamma(\varphi_1 \circ \varphi_2) = F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F = \Gamma(\varphi_1) \circ \Gamma(\varphi_2).$$

So $\mathrm{Aut}(\mathbb{D}) \cong \mathrm{Aut}(\mathbb{H})$. Let $SL_2(\mathbb{R}) = \left\{M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R},\ \det(M) = ad - bc = 1\right\}$, the **special linear group**, and for $M \in SL_2(\mathbb{R})$ let $f_M(z) = \frac{az + b}{cz + d}$.

Theorem 2.4

Every automorphism of $\mathbb{H}$ takes the form f_M for some $M \in SL_2(\mathbb{R})$. Conversely, every map of this form is an automorphism of $\mathbb{H}$.

Proof. Write $G = SL_2(\mathbb{R})$.

Step 1. If $M \in G$, f_M maps $\mathbb{H}$ to itself:

$$\mathrm{Im}(f_M(z)) = \frac{(ad - bc)\,\mathrm{Im}(z)}{|cz + d|^2} = \frac{\mathrm{Im}(z)}{|cz + d|^2} > 0 \qquad (z \in \mathbb{H}). \tag{4}$$

Step 2. $f_M \circ f_{M'} = f_{MM'}$ (a straightforward calculation). Hence each f_M is an automorphism, with holomorphic inverse $f_{M^{-1}}$: $(f_M \circ f_{M^{-1}})(z) = f_I(z) = z$.

Step 3. G acts transitively on $\mathbb{H}$; it suffices to map any $z \in \mathbb{H}$ to i. With $d = 0$ in (4), $\mathrm{Im}(f_M(z)) = \mathrm{Im}(z)/|cz|^2$, and we choose $c \in \mathbb{R}$ making this 1; let $M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$, so $f_{M_1}(z)$ has imaginary part 1. Then translate by $M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $b \in \mathbb{R}$, to bring it to i; $M = M_2 M_1$ works.

Step 4. For $\theta \in \mathbb{R}$, $M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in G$, and $F \circ f_{M_\theta} \circ F^{-1}$ is the rotation of angle -2θ in the disc, since $F \circ f_{M_\theta} = e^{-2i\theta} F(z)$ (easily verified).

Step 5. Let $f \in \mathrm{Aut}(\mathbb{H})$ with $f(\beta) = i$, and $N \in G$ with $f_N(i) = \beta$. Then $g = f \circ f_N$ fixes i, so $F \circ g \circ F^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, hence a rotation; by Step 4, $F \circ g \circ F^{-1} = F \circ f_{M_\theta} \circ F^{-1}$ for some θ. So $g = f_{M_\theta}$ and $f = f_{M_\theta N^{-1}}$. ∎

M and $-M$ give the same f_M; identifying them gives the **projective special linear group** $PSL_2(\mathbb{R})$, which is isomorphic to $\mathrm{Aut}(\mathbb{H})$.

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

If $\Omega = \mathbb{C}$ there is no conformal map $F : \Omega \to \mathbb{D}$ (by Liouville F would be constant); Ω must be connected (as $\mathbb{D}$ is) and simply connected (as $\mathbb{D}$ is). Remarkably, these conditions suffice. A subset of $\mathbb{C}$ is **proper** if it is non-empty and not all of $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem)

Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \to \mathbb{D}$ such that $F(z_0) = 0$ and $F'(z_0) > 0$.

Corollary 3.2

Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.

The corollary follows using $\mathbb{D}$ as an intermediate step. *Uniqueness*: if F, G satisfy the conditions, $H = F \circ G^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, so $H(z) = e^{i\theta} z$; $H'(0) > 0$ forces $e^{i\theta} = 1$, so $F = G$. For existence, consider all injective holomorphic $f : \Omega \to \mathbb{D}$ with $f(z_0) = 0$, and choose one with $|f'(z_0)|$ as large as possible; it will fill out $\mathbb{D}$. We need to extract limits from sequences of functions.

3.2 Montel's theorem

A family $\mathcal{F}$ of holomorphic functions on Ω is **normal** if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on every compact subset of Ω (the limit need not be in $\mathcal{F}$). $\mathcal{F}$ is **uniformly bounded on compact subsets** of Ω if for each compact $K \subset \Omega$ there is $B > 0$ with $|f(z)| \le B$ for $z \in K$, $f \in \mathcal{F}$. $\mathcal{F}$ is **equicontinuous** on a compact K if for every $\epsilon > 0$ there is $\delta > 0$ such that $z, w \in K$, $|z - w| < \delta$ imply $|f(z) - f(w)| < \epsilon$ for all $f \in \mathcal{F}$. (E.g. differentiable functions on $[0, 1]$ with uniformly bounded derivatives are equicontinuous; $f_n(x) = x^n$ on $[0, 1]$ are not.)

Theorem 3.3 (Montel)

Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω. Then:
(i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω.
(ii) $\mathcal{F}$ is a normal family.

Part (i) uses the Cauchy integral formula and contrasts with the real case ($\sin(nx)$ on $(0,1)$ is uniformly bounded but not equicontinuous, with no convergent subsequence on any subinterval). Part (ii) uses only uniform boundedness and equicontinuity (the Arzelà–Ascoli theorem). A sequence $\{K_\ell\}_{\ell=1}^{\infty}$ of compact subsets of Ω is an **exhaustion** if (a) K_ℓ is contained in the interior of $K_{\ell+1}$ for all ℓ, and (b) any compact $K \subset \Omega$ is contained in some K_ℓ; in particular $\Omega = \bigcup_\ell K_\ell$.

Lemma 3.4

Any open set Ω in the complex plane has an exhaustion.

Proof. If Ω is bounded, let K_ℓ be the set of points of Ω at distance $\geq 1/\ell$ from $\partial\Omega$. If Ω is unbounded, require also $|z| \leq \ell$ for $z \in K_\ell$. ■

Proof of Montel's theorem. (i) Let $K \subset \Omega$ be compact and $r > 0$ with $3r < d(K, \partial\Omega)$, so $D_{3r}(z) \subset \Omega$ for $z \in K$. Let $z, w \in K$ with $|z - w| < r$ and γ the boundary circle of $D_{2r}(w)$. By Cauchy's integral formula

$$f(z) - f(w) = \frac{1}{2\pi i}\int_\gamma f(\zeta)\Big[\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big]d\zeta, \qquad \Big|\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big| = \frac{|z - w|}{|\zeta - z||\zeta - w|} \leq \frac{|z - w|}{r^2},$$

since $\zeta \in \gamma$ and $|z - w| < r$. Hence $|f(z) - f(w)| \leq \frac{1}{2\pi} 2\pi(2r)\frac{B|z-w|}{r^2}$, B the uniform bound for $\mathcal{F}$ on the compact set of points at distance $\leq 2r$ from K. So $|f(z) - f(w)| < C|z - w|$ for all $f \in \mathcal{F}$: equicontinuity.

(ii) Let $\{f_n\} \subset \mathcal{F}$ and $K \subset \Omega$ compact. Choose a sequence $\{w_j\}$ dense in Ω. Since $\{f_n\}$ is uniformly bounded, there is a subsequence $\{f_{n,1}\}$ with $f_{n,1}(w_1)$ convergent; from it a subsequence $\{f_{n,2}\}$ with $f_{n,2}(w_2)$ convergent; and so on. The diagonal sequence $g_n = f_{n,n}$ converges at each w_j. It converges uniformly on K: given $\epsilon > 0$, choose δ as in equicontinuity; K is covered by finitely many discs $D_\delta(w_1), \ldots, D_\delta(w_J)$. Choose N with $|g_m(w_j) - g_n(w_j)| < \epsilon$ for $n, m > N$, $j \leq J$. For $z \in K$, $z \in D_\delta(w_j)$ for some $j \leq J$, and

$$|g_n(z) - g_m(z)| \leq |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + |g_m(w_j) - g_m(z)| < 3\epsilon.$$

Finally, with an exhaustion $K_1 \subset K_2 \subset \cdots$, extract $\{g_{n,1}\}$ converging uniformly on K_1, then $\{g_{n,2}\} \subset \{g_{n,1}\}$ on K_2, and so on; $\{g_{n,n}\}$ converges uniformly on every K_ℓ, hence on every compact subset of Ω. ■

Proposition 3.5

If Ω is a connected open subset of $\mathbb{C}$ and $\{f_n\}$ a sequence of injective holomorphic functions on Ω that converges uniformly on every compact subset of Ω to a holomorphic function f, then f is either injective or constant.

Proof. Suppose $f(z_1) = f(z_2)$ with $z_1 \neq z_2$. Let $g_n(z) = f_n(z) - f_n(z_1)$, which has no zero besides z_1, and $g_n \to g = f - f(z_1)$ uniformly on compact sets. If $g \not\equiv 0$, z_2 is an isolated zero of g (Ω is connected), so

$$1 \leq \frac{1}{2\pi i}\int_\gamma \frac{g'(\zeta)}{g(\zeta)}\,d\zeta,$$

γ a small circle about z_2 on and inside which g vanishes only at z_2. Then $1/g_n \to 1/g$ and $g_n' \to g'$ uniformly on γ (Theorem 5.3 of Chapter 2), so $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} \to \frac{1}{2\pi i}\int_\gamma \frac{g'}{g}$. But g_n has no zeros inside γ, so $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} = 0$ for all n, a contradiction. Hence $g \equiv 0$ and f is constant. ■

3.3 Proof of the Riemann mapping theorem

Step 1. Ω is conformally equivalent to an open subset of $\mathbb{D}$ containing 0. Choose $\alpha \notin \Omega$. $z - \alpha$ never vanishes on the simply connected Ω, so there is a holomorphic $f(z) = \log(z-\alpha)$ with $e^{f(z)} = z - \alpha$; in particular f is injective. Fix $w \in \Omega$; then $f(z) \neq f(w) + 2\pi i$ for $z \in \Omega$ (exponentiating would give $z = w$, hence $f(z) = f(w)$). In fact $f(\Omega)$ stays away from $f(w) + 2\pi i$: if $f(z_n) \to f(w) + 2\pi i$, exponentiating gives $z_n \to w$, hence $f(z_n) \to f(w)$, a contradiction. Thus

$$F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$$

is injective and bounded on Ω, so $F : \Omega \to F(\Omega)$ is conformal with $F(\Omega)$ bounded; translating and rescaling gives a conformal map of Ω onto an open subset of $\mathbb{D}$ containing 0.

Step 2. So assume $\Omega \subset \mathbb{D}$ is open with $0 \in \Omega$, and let

$$\mathcal{F} = \{f : \Omega \to \mathbb{D} \text{ holomorphic, injective, and } f(0) = 0\}.$$

$\mathcal{F} \neq \emptyset$ (it contains the identity) and is uniformly bounded. The numbers $|f'(0)|$, $f \in \mathcal{F}$, are uniformly bounded by the Cauchy inequalities on a small disc about 0. Let $s = \sup_{f\in\mathcal{F}} |f'(0)|$ and $f_n \in \mathcal{F}$ with $|f_n'(0)| \to s$. By Montel's theorem a subsequence converges uniformly on compact sets to a holomorphic f on Ω. Since $s \geq 1$ (the identity is in $\mathcal{F}$), f is non-constant, hence injective by Proposition 3.5. By continuity $|f(z)| \leq 1$, and by the maximum modulus principle $|f(z)| < 1$; also $f(0) = 0$. So $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3. f is surjective onto $\mathbb{D}$. Otherwise we construct a function in $\mathcal{F}$ with derivative at 0 greater than s. Suppose $\alpha \in \mathbb{D}$ is not in $f(\Omega)$, and let $\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha} z}$. $U = (\psi_\alpha \circ f)(\Omega)$ is simply connected and does not contain 0, so a square root $g(w) = e^{\frac{1}{2}\log w}$ is defined on U. Let

$$F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f.$$

F is holomorphic, maps 0 to 0 ($g(\psi_\alpha(0)) = g(\alpha)$), maps Ω into $\mathbb{D}$, and is injective (each factor is). So $F \in \mathcal{F}$. With $h(w) = w^2$,

$$f = \psi_\alpha^{-1} \circ h \circ g^{-1} \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F.$$

$\Phi = \psi_\alpha \circ h \circ \psi_{g(\alpha)}$ maps $\mathbb{D}$ into $\mathbb{D}$ with $\Phi(0) = 0$, and is not injective (since h is not). By the last part of the Schwarz lemma, $|\Phi'(0)| < 1$. Since $f'(0) = \Phi'(0)F'(0)$, $|f'(0)| < |F'(0)|$, contradicting the maximality of $|f'(0)|$.

Finally multiply f by a constant of absolute value 1 so that $f'(0) > 0$. ■

Remark

Simple connectivity was used only for the logarithm and the square root. It would suffice to assume Ω proper and *holomorphically simply connected*: $\int_\gamma f(z)\,dz = 0$ for every holomorphic f in Ω and every closed curve $\gamma \subset \Omega$. See Appendix B.

4 Conformal mappings onto polygons

The Riemann mapping theorem gives little insight into the exact form of the map; but for polygons there are nice formulas: the *Schwarz–Christoffel formula* for conformal maps from the disc (or $\mathbb{H}$) to polygons.

4.1 Some examples

Example 1

For $0 < \alpha < 2$, the map $f(z) = z^\alpha$ of $\mathbb{H}$ onto the sector $\{0 < \arg z < \alpha\pi\}$ (Section 1) can be written

$$z^\alpha = f(z) = \int_0^z f'(\zeta)\,d\zeta = \alpha \int_0^z \zeta^{-\beta}\,d\zeta, \qquad \alpha + \beta = 1,$$

integrated along any path in the (closed) upper half-plane. $\zeta^{-\beta}$ is integrable near 0 since $\beta < 1$, so $f(0) = 0$. For $z = x > 0$, $f'(x) = \alpha x^{\alpha-1} > 0$ and is not integrable at ∞, so f maps $[0,\infty)$ onto $[0,\infty)$. For $x < 0$, $f'(x) = \alpha|x|^{\alpha-1}e^{i\pi(\alpha-1)} = -\alpha|x|^{\alpha-1}e^{i\pi\alpha}$, so f maps $(-\infty, 0]$ onto $(\infty e^{i\pi\alpha}, 0]$.

Example 2

For $z \in \mathbb{H}$ let $f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}}$, with the branch of $(1-\zeta^2)^{1/2}$ holomorphic in $\mathbb{H}$ and positive for $-1 < \zeta < 1$; then $(1-\zeta^2)^{-1/2} = -i(\zeta^2-1)^{-1/2}$ for $\zeta > 1$ (and the analogous formula for $\zeta < -1$). f maps $\mathbb{R}$ onto the boundary of the half-strip $\{-\pi/2 < \mathrm{Re} < \pi/2,\ \mathrm{Im} > 0\}$: $f(\pm 1) = \pm\pi/2$ and $f'(x) > 0$ on $(-1,1)$, so $[-1,1] \mapsto [-\pi/2, \pi/2]$; for $x > 1$, $f(x) = \frac{\pi}{2} + i\int_1^x \frac{dx}{(x^2-1)^{1/2}}$ and $\int_1^\infty \frac{dx}{(x^2-1)^{1/2}} = \infty$, so $[1,\infty)$ goes onto the vertical half-line above $\pi/2$; similarly for $(-\infty, -1]$. In fact f is the inverse of $\sin z$ (Example 8), so f maps $\mathbb{H}$ conformally onto the half-strip.

Example 3

For fixed $0 < k < 1$, let

$$f(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}, \qquad z \in \mathbb{H},$$

the branch positive for $-1 < \zeta < 1$. Such integrals are **elliptic integrals** (variants arise in the arc length of an ellipse). With

$$K = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K' = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}},$$

f maps $\mathbb{R}$ onto the rectangle with vertices $-K$, K, $K+iK'$, $-K+iK'$. Divide $\mathbb{R}$ by $-1/k, -1, 1, 1/k$. $f(\pm 1) = \pm K$ and $f' > 0$ on $(-1,1)$, so $[-1,1] \mapsto [-K, K]$. For $1 < x < 1/k$, $f(x) = K + i\int_1^x \frac{d\xi}{[(\xi^2-1)(1-k^2\xi^2)]^{1/2}}$, so $[1, 1/k] \mapsto [K, K+iK']$; similarly $[-1/k, -1] \mapsto [-K+iK', -K]$. For $x > 1/k$, $f'(x) = -[(x^2-1)(k^2x^2-1)]^{-1/2}$, so $f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}}$, and since $\int_{1/k}^\infty \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}} = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}} = K$ (substitute $x = 1/ku$), $[1/k, \infty)$ goes onto $[K+iK', iK')$; similarly $(-\infty, -1/k]$ onto $[-K+iK', iK')$. The point at infinity goes to the midpoint iK' of the upper side.

Two problems arise: generalize these examples (the Schwarz–Christoffel integral maps $\mathbb{R}$ to a polygonal line), and show that the map is conformal from $\mathbb{H}$ onto the interior of the polygon.

4.2 The Schwarz–Christoffel integral

Definition (Schwarz–Christoffel integral)

For real $A_1 < A_2 < \cdots < A_n$ and exponents $\beta_k < 1$ with $1 < \sum_{k=1}^n \beta_k$, let

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}, \tag{5}$$

where $(z - A_k)^{\beta_k}$ is the branch in the plane slit along $\{A_k + iy : y \le 0\}$ that is positive for real $x > A_k$; thus $(x - A_k)^{\beta_k} = |x - A_k|^{\beta_k} e^{i\pi\beta_k}$ for real $x < A_k$.

The plane slit along the rays $\bigcup_k \{A_k + iy : y \le 0\}$ is simply connected, so S is holomorphic there. Since $\beta_k < 1$, the singularities are integrable, and S is continuous up to $\mathbb{R}$, including the A_k; the integral may be taken along any path avoiding the open slits. Since $\prod_k |\zeta - A_k|^{-\beta_k} \le c|\zeta|^{-\sum \beta_k}$ for large $|\zeta|$ and $\sum \beta_k > 1$, the integral converges at infinity, and by Cauchy's theorem $\lim_{r\to\infty} S(re^{i\theta})$ exists and is independent of $0 \le \theta \le \pi$; call it a_∞, and let $a_k = S(A_k)$. (The cases $\sum \beta_k \le 1$ of Examples 1–2 are excluded; a modification handles them, but then S is unbounded in $\mathbb{H}$.)

Proposition 4.1

Suppose $S(z)$ is given by (5).
(i) If $\sum_{k=1}^n \beta_k = 2$, and p denotes the polygon whose vertices are given (in order) by $a_1, \ldots, a_n$, then S maps the real axis onto $p - \{a_\infty\}$. The point a_∞ lies on the segment $[a_n, a_1]$ and is the image of the point at infinity. Moreover, the (interior) angle at the vertex a_k is $\alpha_k \pi$ where $\alpha_k = 1 - \beta_k$.
(ii) There is a similar conclusion when $1 < \sum_{k=1}^n \beta_k < 2$, except now the image of the extended line is the polygon of $n+1$ sides with vertices $a_1, a_2, \ldots, a_n, a_\infty$. The angle at the vertex a_∞ is $\alpha_\infty \pi$ with $\alpha_\infty = 1 - \beta_\infty$, where $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.

Proof. Let $\sum \beta_k = 2$. For $A_k < x < A_{k+1}$,

$$S'(x) = \prod_{j \le k} (x - A_j)^{-\beta_j} \prod_{j > k} (x - A_j)^{-\beta_j}, \qquad \arg S'(x) = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

constant on (A_k, A_{k+1}). Since $S(x) = S(A_k) + \int_{A_k}^x S'(y)\, dy$, as x goes from A_k to A_{k+1}, $S(x)$ moves along the straight segment $[a_k, a_{k+1}]$ (where $[z, w] = \{(1-t)z + tw : t \in [0,1]\}$), which makes angle $-\pi \sum_{j>k} \beta_j$ with the real axis. For $x > A_n$, $S'(x) > 0$; for $x < A_1$, $\arg S'(x) = -\pi \sum_k \beta_k = -2\pi$, so $S'(x) > 0$ again. Thus on $[A_n, \infty)$, S moves parallel to the real axis from a_n to a_∞, and on $(-\infty, A_1]$ from a_∞ to a_1; $[a_n, a_\infty) \cup (a_\infty, a_1] = [a_n, a_1] - \{a_\infty\}$. The angle of $[a_k, a_{k+1}]$ exceeds that of $[a_{k-1}, a_k]$ by $\pi\beta_k$, so the interior angle at a_k is $\pi - \pi\beta_k = \alpha_k \pi$. The case $1 < \sum \beta_k < 2$ is similar. ■

This does not settle the problem:

(1) for general n and A_k the image polygon need not be simple, nor S conformal on $\mathbb{H}$;

(2) it does not show that for a given simply connected region P bounded by a polygon, S (with suitable A_k and simple modifications) is a conformal map of $\mathbb{H}$ onto P. That is the case, as we now prove.

4.3 Boundary behavior

A **polygonal region** P is a bounded simply connected open set whose boundary is a closed polygonal line p (a *polygon*). For a conformal $f : U \to f(U)$,

$$\text{Area}(f(U)) = \iint_U |f'(z)|^2 \, dx\, dy,$$

since the Jacobian of $w = f(z)$ is $|f'(z)|^2$ (Proposition 2.3 of Chapter 1).

Theorem 4.2

If $F : \mathbb{D} \to P$ is a conformal map, then F extends to a continuous bijection from the closure $\overline{\mathbb{D}}$ of the disc to the closure $\overline{P}$ of the polygonal region. In particular, F gives rise to a bijection from the boundary of the disc to the boundary polygon p.

Lemma 4.3

For each $0 < r < 1/2$, let C_r denote the circle centered at z_0 (a point of the unit circle) of radius r. Suppose that for all sufficiently small r we are given two points z_r and z_r' in the unit disc that also lie on C_r. If we let $\rho(r) = |F(z_r) - F(z_r')|$, then there exists a sequence $\{r_n\}$ of radii that tends to zero, and such that $\lim_{n\to\infty} \rho(r_n) = 0$.

Proof. If not, there are $c > 0$, $0 < R < 1/2$ with $c \le \rho(r)$ for $0 < r \le R$. Now $F(z_r) - F(z_r') = \int_\gamma F'(\zeta)\, d\zeta$, γ the arc of C_r in $\mathbb{D}$ joining z_r, z_r'; parametrizing it by $z_0 + re^{i\theta}$, $\theta_1(r) \le \theta \le \theta_2(r)$,

$$\rho(r) \le \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)| r \, d\theta \le \Big(\int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r \, d\theta \Big)^{1/2} \Big(\int_{\theta_1(r)}^{\theta_2(r)} r \, d\theta \Big)^{1/2}$$

by Cauchy–Schwarz. Squaring and dividing by r, $\frac{\rho(r)^2}{r} \le 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r \, d\theta$. Integrating from 0 to R,

$$c^2 \int_0^R \frac{dr}{r} \le 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r \, d\theta \, dr \le 2\pi \iint_{\mathbb{D}} |F'(z)|^2 \, dx\, dy = 2\pi \, \text{Area}(P).$$

The left side is infinite and the right side finite—a contradiction. ∎

Lemma 4.4

Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.

Proof. If not, there are sequences $z_n, z_n' \to z_0$ in $\mathbb{D}$ with $F(z_n) \to \zeta$, $F(z_n') \to \zeta'$, distinct points of $\overline{P}$; since F is conformal, they lie on p. Choose disjoint discs D, D' about ζ, ζ' at distance $d > 0$. For large n, $F(z_n) \in D$, $F(z_n') \in D'$, so there are continuous curves $\Lambda \subset D \cap P$, $\Lambda' \subset D' \cap P$ (images of continuous maps of a closed interval) containing $F(z_n)$, $F(z_n')$ for all large n, with end-points ζ, ζ'. Then $\lambda = F^{-1}(\Lambda)$ and $\lambda' = F^{-1}(\Lambda')$ are continuous curves in $\mathbb{D}$, containing infinitely many z_n, resp. z_n', which tend to z_0. By continuity, for all small r the circle C_r meets λ and λ' at points z_r, z_r', and $|F(z_r) - F(z_r')| > d$, contradicting Lemma 4.3. ∎

Lemma 4.5

The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.

Proof. Define $F(z_0) = \lim_{z\to z_0} F(z)$ for $|z_0| = 1$ (Lemma 4.4). Given ϵ, there is δ with $|F(z) - F(z_0)| < \epsilon$ for $z \in \mathbb{D}$, $|z - z_0| < \delta$. If $|z| = 1$ and $|z - z_0| < \delta$, choose $w \in \mathbb{D}$ with $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$; then $|F(z) - F(z_0)| \le |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon$. ■

Proof of Theorem 4.2. F extends continuously to $\overline{\mathbb{D}} \to \overline{P}$. The argument applies to the inverse G of F: the key geometric property used was that for $z_0 \in \partial\mathbb{D}$ and a small circle C about z_0, $C \cap \mathbb{D}$ is an arc; this holds at every boundary point of P. So G extends continuously to $\overline{P} \to \overline{\mathbb{D}}$. If $z \in \overline{\mathbb{D}}$ and $z_k \to z$ in $\mathbb{D}$, then $G(F(z_k)) = z_k$, and by continuity $G(F(z)) = z$; similarly $F(G(w)) = w$ for $w \in \overline{P}$. ■

4.4 The mapping formula

Let P be a polygonal region bounded by p with consecutive vertices $a_1, \dots, a_n$, $n \ge 3$, interior angle $\alpha_k\pi$ at a_k, and exterior angle $\beta_k\pi$ with $\alpha_k + \beta_k = 1$; a simple geometric argument gives $\sum_{k=1}^{n} \beta_k = 2$. The correspondence $w = (i - z)/(i + z)$ between $\mathbb{H}$ and $\mathbb{D}$ sends the point at infinity to $w = -1$, so a conformal map of $\mathbb{H}$ onto P (which exists by the Riemann mapping theorem) extends to a continuous bijection of $\mathbb{R} \cup \{\infty\}$ onto p. Let F be such a map, and first assume no vertex corresponds to ∞: $F(A_k) = a_k$ for real A_k. Since F is continuous and injective and the vertices are consecutive, after relabeling $A_1 < A_2 < \cdots < A_n$, and F maps $[A_k, A_{k+1}]$ onto $[a_k, a_{k+1}]$ and $(-\infty, A_1] \cup [A_n, \infty)$ onto $[a_n, a_1]$.

Theorem 4.6 (Schwarz–Christoffel)

There exist complex numbers c_1 and c_2 so that the conformal map F of $\mathbb{H}$ to P is given by $F(z) = c_1 S(z) + c_2$, where S is the Schwarz–Christoffel integral introduced in Section 4.2.

Proof. Let $1 < k < n$. F maps $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$ onto two segments meeting at $a_k = F(A_k)$ at angle $\alpha_k\pi$. Choosing a branch of the logarithm, define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

on the half-strip in $\mathbb{H}$ between $\mathrm{Re}(z) = A_{k-1}$ and $\mathrm{Re}(z) = A_{k+1}$. h_k is continuous up to (A_{k-1}, A_{k+1}) and maps $[A_{k-1}, A_{k+1}]$ onto a straight segment L_k, with $A_k \mapsto 0$. By the Schwarz reflection principle, h_k continues analytically to the whole strip $A_{k-1} < \mathrm{Re}(z) < A_{k+1}$. h_k' never vanishes there: in the upper half-strip,

$$\frac{F'(z)}{F(z) - F(A_k)} = \frac{1}{\alpha_k}\frac{h_k'(z)}{h_k(z)},$$

and $F'(z) \ne 0$ (Proposition 1.1), so $h_k'(z) \ne 0$; by reflection the same holds in the lower half-strip. For $A_{k-1} < x < A_{k+1}$, the image under h_k of a small half-disc about x in $\mathbb{H}$ lies on one side of L_k; h_k is injective up to L_k (since F is), and by the symmetry of reflection h_k is injective on the whole disc about x, so $h_k'(x) \ne 0$.

Since $F = a_k + h_k^{\alpha_k}$, $F' = \alpha_k h_k^{\alpha_k - 1} h_k'$ and $F'' = \alpha_k(\alpha_k - 1)h_k^{\alpha_k - 2}(h_k')^2 + \alpha_k h_k^{\alpha_k - 1} h_k''$; as $h_k' \ne 0$ and h_k has a simple zero at A_k,

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

E_k holomorphic in the strip $A_{k-1} < \mathrm{Re}(z) < A_{k+1}$. Similarly $\frac{F''}{F'} = \frac{-\beta_1}{z - A_1} + E_1$ in $-\infty < \mathrm{Re}(z) < A_2$ and $\frac{F''}{F'} = \frac{-\beta_n}{z - A_n} + E_n$ in $A_{n-1} < \mathrm{Re}(z) < \infty$. Moreover F continues across $(-\infty, A_1) \cup (A_n, \infty)$ by reflection (the image is a straight segment), so F continues to $|z| > R$ for large R; since F maps $\mathbb{H}$ into a bounded region, the continuation is bounded, hence holomorphic at infinity: $F(z) = c_0 + \frac{c_1}{z} + \frac{c_2}{z^2} + \cdots$. Differentiating, F''/F' decays like $1/z$ at infinity. Since the strips overlap and cover $\mathbb{C}$,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^{n} \frac{\beta_k}{z - A_k}$$

is entire and vanishes at infinity; by Liouville it is zero. With $Q(z) = (z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$, $\frac{Q'}{Q} = -\sum_k \frac{\beta_k}{z - A_k} = \frac{F''}{F'}$, so $\frac{d}{dz}\left(\frac{F'(z)}{Q(z)}\right) = 0$ and $F' = cQ$. A final integration yields the theorem. ■

Theorem 4.7

If F is a conformal map from the upper half-plane to the polygonal region P and maps the points $A_1, \ldots, A_{n-1}, \infty$ to the vertices of p, then there exist constants C_1 and C_2 such that

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

So the formula is obtained by deleting the last factor of the Schwarz–Christoffel integral.

Proof. After a translation assume $A_j \neq 0$ for $j = 1, \ldots, n-1$. Choose a real $A_n^* > 0$ (different from the others; [Supplement]) and let $\Phi(z) = A_n^* - \frac{1}{z}$, an automorphism of $\mathbb{H}$. Let $A_k^* = \Phi(A_k)$, $k \leq n-1$, and note $A_n^* = \Phi(\infty)$. Then $(F \circ \Phi^{-1})(A_k^*) = a_k$ for $k = 1, \ldots, n$, and by Theorem 4.6

$$(F \circ \Phi^{-1})(z') = C_1' \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2'.$$

Change variables $\zeta = \Phi(w)$, $d\zeta = dw/w^2$; since $\zeta - A_n^* = -1/w$ and $\zeta - A_k^* = (w(A_n^* - A_k^*) - 1)/w$, and $2 = \beta_1 + \cdots + \beta_n$, the integrand becomes $C \prod_{k<n}(w - 1/(A_n^* - A_k^*))^{-\beta_k}$ up to a constant factor. Since $1/(A_n^* - A_k^*) = A_k$, setting $\Phi^{-1}(z') = z$ gives the formula. ■

4.5 Return to elliptic integrals

Consider again $I(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}$, $0 < k < 1$, which maps $\mathbb{R}$ onto the rectangle R with vertices $-K, K, K+iK', -K+iK'$ (Example 3). We show I is a conformal map of $\mathbb{H}$ onto the interior of R. By Theorem 4.6 there is a conformal map F of $\mathbb{H}$ onto R mapping four real points to the vertices; composing with an automorphism of $\mathbb{H}$ we may assume F maps $-1, 0, 1$ to $-K, 0, K$. Let $1/\ell$ $(0 < \ell < 1)$ be the real point mapped to $K + iK'$. Then $F(-1/\ell) = -K + iK'$: $\tilde{F}(z) = -\overline{F(-\overline{z})}$ is, by the symmetry of R, also a conformal map of $\mathbb{H}$ onto R with $\tilde{F}(0) = 0$, $\tilde{F}(\pm 1) = \pm K$; so $F^{-1} \circ \tilde{F}$ is an automorphism of $\mathbb{H}$ fixing $-1, 0, 1$, hence the identity, and $F(-1/\ell) = -\overline{F(1/\ell)} = -K + iK'$. By Theorem 4.6,

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-\ell^2\zeta^2)]^{1/2}} + c_2.$$

$z = 0$ gives $c_2 = 0$; $z = 1$ and $z = 1/\ell$ give $K(k) = c_1 K(\ell)$ and $K'(k) = c_1 K'(\ell)$, where $K(k) = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}$, $K'(k) = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}}$. $K(k)$ is strictly increasing in $k \in (0,1)$, and a change of variables gives $K'(k) = K(\tilde{k})$ with $\tilde{k}^2 = 1 - k^2$, $\tilde{k} > 0$, so K' is strictly decreasing. Hence $K(k)/K'(k)$ is strictly increasing; since $K(k)/K'(k) = K(\ell)/K'(\ell)$, $k = \ell$ and $c_1 = 1$. Thus $I = F$ is conformal.

Passing to the inverse function gives basic insight: let $z \mapsto \operatorname{sn}(z)$ be the inverse of $z \mapsto I(z)$ (Jacobi's notation, by analogy with $\sin z$). It maps the closed rectangle $R_0 = R$ onto the closed upper half-plane. Reflect successively along the lower sides to get rectangles $R_1, R_2, \ldots$: by the reflection principle, $\operatorname{sn}(z) = \overline{\operatorname{sn}(\overline{z})}$ for $z \in R_1$, then $\operatorname{sn}(z) = \operatorname{sn}(-iK' + \overline{z})$ for $z \in R_2$, and so on; this extends sn to the strip $-K < \operatorname{Re}(z) < K$ with $\operatorname{sn}(z) = \operatorname{sn}(z + 2iK')$. Reflecting also in horizontal rows of rectangles, sn continues to all of $\mathbb{C}$ with $\operatorname{sn}(z) = \operatorname{sn}(z + 4K)$. Thus sn is *doubly periodic*, with periods $4K$ and $2iK'$ (and meromorphic, with two simple poles in a period parallelogram)—the subject of Chapter 9.

Chapter 9. An Introduction to Elliptic Functions

Overview

Elliptic functions grew out of the study of *elliptic integrals* $\int R(x, \sqrt{P(x)})\,dx$, R rational and P a polynomial of degree three or four (degree two reduces to the circular functions), which arose in computing the arc length of an ellipse or lemniscate. Their early study centered on transformation properties and an inherent double periodicity (cf. the map of $\mathbb{H}$ onto a rectangle, Section 4). Jacobi began the systematic study of doubly periodic functions (*elliptic functions*), with theta functions playing a decisive role; Weierstrass's approach, based on his function $\wp$, is simpler in its first steps, and we sketch it here, reaching the *Eisenstein series* and their expression via divisor functions, a link with number theory. Theta functions and further number-theoretic applications follow in Chapter 10.

1 Elliptic functions

We consider meromorphic f on $\mathbb{C}$ with two periods: non-zero ω_1, ω_2 with $f(z+\omega_1) = f(z)$ and $f(z+\omega_2) = f(z)$ for all z; f is **doubly periodic**. If $\omega_2/\omega_1 \in \mathbb{R}$, f is either simply periodic (quotient rational) or constant (quotient irrational), so we assume ω_1, ω_2 are linearly independent over $\mathbb{R}$. *Normalization*: let $\tau = \omega_2/\omega_1$; since τ and $1/\tau$ have imaginary parts of opposite signs, after possibly interchanging ω_1, ω_2 we may assume $\operatorname{Im}(\tau) > 0$. f has periods ω_1, ω_2 iff $F(z) = f(\omega_1 z)$ has periods 1 and τ, and F is meromorphic iff f is. So we assume f is meromorphic on $\mathbb{C}$ with periods 1 and τ, $\operatorname{Im}(\tau) > 0$. Then

$$f(z+n+m\tau) = f(z) \qquad \text{for all integers } n, m \text{ and all } z \in \mathbb{C}, \tag{1}$$

and we consider the **lattice** $\Lambda = \{n+m\tau : n, m \in \mathbb{Z}\}$ **generated** by 1 and τ, and the **fundamental parallelogram**

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau,\ 0 \le a < 1,\ 0 \le b < 1\}.$$

z and w are **congruent modulo** Λ, $z \sim w$, if $z - w \in \Lambda$; by (1), $f(z) = f(w)$ when $z \sim w$. Every $z \in \mathbb{C}$ is congruent to a unique point of P_0: write $z = a + b\tau$ with $a, b \in \mathbb{R}$ (1 and τ form a real basis of $\mathbb{C}$), let n, m be the greatest integers $\le a$, $\le b$, and $w = z - n - m\tau = (a-n) + (b-m)\tau \in P_0$. If $w = a + b\tau$ and $w' = a' + b'\tau$ in P_0 are congruent, then $a - a'$, $b - b'$ are integers in $(-1, 1)$, hence 0. A **period parallelogram** is any translate $P = P_0 + h$; applying the above to $z - h$, every point is congruent to a unique point of P. Moreover

$$\mathbb{C} = \bigcup_{n,m \in \mathbb{Z}} (n + m\tau + P_0), \tag{2}$$

a disjoint union.

Proposition 1.1

Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ. Then:
(i) Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram.
(ii) Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram.
(iii) The lattice Λ provides a disjoint covering of the complex plane, in the sense of (2).
(iv) The function f is completely determined by its values in any period parallelogram.

1.1 Liouville's theorems

Theorem 1.2

An entire doubly periodic function is constant.

Proof. It is determined by its values on P_0, whose closure is compact; so it is bounded on $\mathbb{C}$, hence constant by Liouville's theorem. ■

A non-constant doubly periodic meromorphic function is an **elliptic function**. It has finitely many zeros and poles in any period parallelogram (possibly some on ∂P_0); we count them with multiplicities.

Theorem 1.3

The total number of poles of an elliptic function in P_0 is always ≥ 2.

So f cannot have just one simple pole (a single double pole is allowed).

Proof. Suppose first f has no poles on ∂P_0. By the residue theorem $\int_{\partial P_0} f(z)\,dz = 2\pi i \sum \operatorname{res} f$, and the integral is 0 by periodicity:

$$\int_{\partial P_0} f = \int_0^1 f + \int_1^{1+\tau} f + \int_{1+\tau}^{\tau} f + \int_{\tau}^{0} f,$$

and opposite sides cancel; e.g.

$$\int_0^1 f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz = \int_0^1 f(z)\,dz + \int_1^0 f(s+\tau)\,ds = \int_0^1 f(z)\,dz - \int_0^1 f(s)\,ds = 0.$$

So $\sum \operatorname{res} f = 0$, and f (having a pole) must have at least two poles in P_0. If f has a pole on ∂P_0, choose a small h with no poles on $\partial(h + P_0)$ and argue with $P = h + P_0$. ■

The total number of poles (with multiplicities) of an elliptic function is its **order**.

Theorem 1.4

Every elliptic function of order m has m zeros in P_0.

Proof. If f has no zeros or poles on ∂P_0, the argument principle gives $\int_{\partial P_0} \frac{f'}{f} = 2\pi i(N_z - N_p)$, and by periodicity (as above) the integral is 0, so $N_z = N_p$. Otherwise apply the argument to a translate of P_0. ■

Consequently, for every $c \in \mathbb{C}$, $f(z) = c$ has as many solutions in P_0 as the order of f ($f - c$ is elliptic with the same poles).

1.2 The Weierstrass $\wp$ function

An elliptic function of order two. We construct an elliptic function whose only singularities are double poles at the lattice points. For one period, $\sum_{n=-\infty}^{\infty} \frac{1}{z+n}$ is unchanged under $z \mapsto z+1$ with poles at the integers but is not absolutely convergent; summing symmetrically,

$$F(z) = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \Big[\frac{1}{z+n} + \frac{1}{z-n}\Big],$$

the bracket being $O(1/n^2)$; $F(z) = \pi \cot \pi z$ (Chapter 5). Alternatively, write it as $\frac{1}{z} + \sum_{n\neq 0} \big[\frac{1}{z+n} - \frac{1}{n}\big]$, absolutely convergent as $\frac{1}{z+n} - \frac{1}{n} = O(1/n^2)$, and equal to the former since $\big[\frac{1}{z+n} - \frac{1}{n}\big] + \big[\frac{1}{z-n} -$

$\frac{1}{-n}] = \frac{1}{z+n} + \frac{1}{z-n}$. Mimicking the second way: $\sum_{\omega\in\Lambda} \frac{1}{(z+\omega)^2}$ does not converge absolutely, so with $\Lambda^* = \Lambda - \{0\}$ consider

$$\frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big], \qquad \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2-2z\omega}{(z+\omega)^2\omega^2} = O\Big(\frac{1}{|\omega|^3}\Big) \text{ as } |\omega|\to\infty.$$

Lemma 1.5

The two series

$$\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} \qquad \text{and} \qquad \sum_{n+m\tau\in\Lambda^*} \frac{1}{|n+m\tau|^r}$$

converge if $r > 2$.

(By the Note at the end of Chapter 7, absolute convergence of a double series does not depend on the order of summation; we sum first in m, then in n.)

Proof. For $n \neq 0$, by integral comparison ($1/k^r \le \int_{k-1}^k dx/x^r$),

$$\sum_{m\in\mathbb{Z}} \frac{1}{(|n|+|m|)^r} = \frac{1}{|n|^r} + 2\sum_{k\ge|n|+1} \frac{1}{k^r} \le \frac{1}{|n|^r} + 2\int_{|n|}^\infty \frac{dx}{x^r} \le \frac{1}{|n|^r} + \frac{C}{|n|^{r-1}},$$

so for $r > 2$, $\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} = \sum_{|m|\neq 0} \frac{1}{|m|^r} + \sum_{|n|\neq 0}\sum_m \frac{1}{(|n|+|m|)^r} < \infty$. For the second series it suffices that $|n|+|m| \le c|n+m\tau|$. Write $A \approx B$ if $A \lesssim B$ and $B \lesssim A$ ($x \lesssim y$ meaning $x \le ay$ for some $a > 0$). For positive A, B, $(A^2+B^2)^{1/2} \approx A + B$ (squaring gives one direction, and $A, B \le (A^2+B^2)^{1/2}$ the other). With $\tau = s + it$, $t > 0$,

$$|n+m\tau| = [(n+ms)^2 + (mt)^2]^{1/2} \approx |n+ms| + |mt| \approx |n+ms| + |m|,$$

and $|n+ms| + |m| \approx |n| + |m|$, considering separately $|n| \ge 2|m||s|$ and $|n| < 2|m||s|$. ∎

Remark

The proof shows that for $r > 2$ the series $\sum|n+m\tau|^{-r}$ converges uniformly in every half-plane $\mathrm{Im}(\tau) \ge \delta > 0$. For $r = 2$ it diverges.

The **Weierstrass $\wp$ function** is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big] = \frac{1}{z^2} + \sum_{(n,m)\neq(0,0)} \Big[\frac{1}{(z+n+m\tau)^2} - \frac{1}{(n+m\tau)^2}\Big].$$

It is meromorphic with double poles at the lattice points: for $|z| < R$ split the sum into $|\omega| \le 2R$ and $|\omega| > 2R$; the terms of the second sum are $O(1/|\omega|^3)$ uniformly, so by Lemma 1.5 it is holomorphic in $|z| < R$, and the finite first sum exhibits the double poles.

Theorem 1.6

The function $\wp$ is an elliptic function that has periods 1 and τ, and double poles at the lattice points.

Proof. It remains to prove periodicity. Differentiating termwise,

$$\wp'(z) = -2\sum_{n,m\in\mathbb{Z}} \frac{1}{(z+n+m\tau)^3},$$

which converges absolutely for $z \notin \Lambda$ ($r = 3$ in Lemma 1.5) and is clearly periodic with periods 1, τ, the terms $1/\omega^2$ having disappeared. So $\wp(z+1) = \wp(z) + a$ and $\wp(z+\tau) = \wp(z) + b$. $\wp$ is even (replace the sum over ω by the sum over $-\omega$), so $\wp(-1/2) = \wp(1/2)$ and $\wp(-\tau/2) = \wp(\tau/2)$; setting $z = -1/2$ and $z = -\tau/2$ gives $a = b = 0$. ∎

Properties of $\wp$. $\wp$ is even, so $\wp'$ is odd; with periodicity,

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\Big(\frac{1+\tau}{2}\Big) = 0,$$

e.g. $\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2+1) = -\wp'(1/2)$. Since $\wp'$ is elliptic of order 3, the **half-periods** $1/2$, $\tau/2$, $(1+\tau)/2$ are its only roots in P_0, each of multiplicity 1. Let

$$\wp(1/2) = e_1, \qquad \wp(\tau/2) = e_2, \qquad \wp\Big(\frac{1+\tau}{2}\Big) = e_3.$$

Then $\wp(z) = e_1$ has a double root at $1/2$, and since $\wp$ has order 2, no other solutions in P_0; similarly for e_2, e_3. In particular e_1, e_2, e_3 are distinct (else $\wp$ would take a value four times).

Theorem 1.7

The function $(\wp')^2$ is the cubic polynomial in $\wp$

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. $F(z) = (\wp(z)-e_1)(\wp(z)-e_2)(\wp(z)-e_3)$ has, in P_0, only double roots at $1/2$, $\tau/2$, $(1+\tau)/2$; so does $(\wp')^2$. Both have poles of order 6 at the lattice points. So $(\wp')^2/F$ is holomorphic and doubly periodic, hence constant. Near 0, $\wp(z) = \frac{1}{z^2} + \cdots$ and $\wp'(z) = \frac{-2}{z^3} + \cdots$, so the constant is 4. ∎

$\wp$ is universal:

Theorem 1.8

Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.

Lemma 1.9

Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.

Proof. A zero or pole of the even function F at 0 has even order, so for some integer m, $\wp^m F$ has no zero or pole at the lattice points; we may assume F itself has none. We construct a doubly periodic G with the same zeros and poles as F, recalling that $\wp(z)-\wp(a)$ has a single zero of order 2 if a is a half-period, and two distinct zeros a, $-a$ otherwise. If a is a zero of F, so is $-a$ (F is even); $a \sim -a$ iff a is a half-period, in which case the zero has even order. So if $a_1, -a_1, \ldots, a_m, -a_m$ (with multiplicities; for a half-period a_j, each of a_j, $-a_j$ carries half the multiplicity) are all the zeros of F in a period parallelogram, $[\wp(z) - \wp(a_1)]\cdots[\wp(z) - \wp(a_m)]$ has precisely the same roots as F. Similarly with the poles $b_1, -b_1, \ldots, b_m, -b_m$,

$$G(z) = \frac{[\wp(z) - \wp(a_1)]\cdots[\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)]\cdots[\wp(z) - \wp(b_m)]}$$

is periodic with the same zeros and poles as F. So F/G is holomorphic and doubly periodic, hence constant. ∎

Proof of Theorem 1.8. Write $f = f_{\text{even}} + f_{\text{odd}}$ with $f_{\text{even}}(z) = \frac{f(z)+f(-z)}{2}$, $f_{\text{odd}}(z) = \frac{f(z)-f(-z)}{2}$. Since $\wp'$ is odd, $f_{\text{odd}}/\wp'$ is even; apply the lemma to f_{even} and $f_{\text{odd}}/\wp'$. ∎

2 The modular character of elliptic functions and Eisenstein series

We now study the dependence on τ and write $\wp_\tau$. Since $1, \tau$ and $1, \tau+1$ generate the same lattice, $\wp_\tau = \wp_{\tau+1}$. Since $\tau = \omega_2/\omega_1$, $-1/\tau = -\omega_1/\omega_2$ (with $\operatorname{Im}(-1/\tau) > 0$) corresponds essentially to

interchanging the periods, and indeed $\wp_{-1/\tau}(z) = \tau^2\wp_\tau(\tau z)$. So we consider the group of transformations of the upper half-plane generated by $\tau \mapsto \tau+1$ and $\tau \mapsto -1/\tau$, the **modular group**; quantities attached to $\wp_\tau$ should reflect these transformations.

2.1 Eisenstein series

The **Eisenstein series of order** k is

$$E_k(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^k} = \sum_{\omega\in\Lambda^*} \frac{1}{\omega^k},$$

for integers $k \geq 3$ and $\mathrm{Im}(\tau) > 0$.

Theorem 2.1

Eisenstein series have the following properties:
(i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane.
(ii) $E_k(\tau) = 0$ if k is odd.
(iii) $E_k(\tau)$ satisfies the following transformation relations: $E_k(\tau+1) = E_k(\tau)$ and $E_k(\tau) = \tau^{-k}E_k(-1/\tau)$.

Property (iii) is the **modular character** of the Eisenstein series.

Proof. (i) By Lemma 1.5 and the remark after it, the series converges absolutely and uniformly in every half-plane $\mathrm{Im}(\tau) \geq \delta > 0$ for $k \geq 3$.

(ii) Replacing (n,m) by $(-n,-m)$ shows $E_k = -E_k$ for odd k.

(iii) $n + m(\tau+1) = (n+m) + m\tau$, and we may replace $n+m$ by n. Also $(n+m(-1/\tau))^k = \tau^{-k}(n\tau - m)^k$, and rearranging, replacing $(-m,n)$ by (n,m), gives the second relation. ∎

(Because of (ii), some authors define the Eisenstein series as $\sum 1/(n+m\tau)^{2k}$, possibly with a constant factor.)

Theorem 2.2

For z near 0, we have

$$\wp(z) = \frac{1}{z^2} + 3E_4z^2 + 5E_6z^4 + \cdots = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k}.$$

Proof. Replacing ω by $-\omega$, $\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*}\left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right]$. Differentiating the geometric series, $\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty}(\ell+1)w^\ell$ for $|w|<1$, so for small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2}\sum_{\ell=0}^{\infty}(\ell+1)\Big(\frac{z}{\omega}\Big)^\ell = \frac{1}{\omega^2} + \sum_{\ell=1}^{\infty}(\ell+1)\frac{z^\ell}{\omega^{\ell+2}}.$$

Hence

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)\Big(\sum_{\omega\in\Lambda^*}\frac{1}{\omega^{\ell+2}}\Big)z^\ell = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)E_{\ell+2}z^\ell = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k},$$

since $E_{\ell+2} = 0$ for odd ℓ. ∎

Hence near 0,

$$\wp'(z) = \frac{-2}{z^3} + 6E_4z + 20E_6z^3 + \cdots,$$

$$(\wp'(z))^2 = \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \qquad (\wp(z))^3 = \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots.$$

So $(\wp')^2 - 4\wp^3 + 60E_4\wp + 140E_6$ is holomorphic near 0 and vanishes there; being doubly periodic, it is constant (Theorem 1.2), hence 0:

Corollary 2.3

If $g_2 = 60E_4$ and $g_3 = 140E_6$, then $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

This is another version of Theorem 1.7, and expresses the symmetric functions of the e_j via Eisenstein series.

2.2 Eisenstein series and divisor functions

Consider the Fourier expansion of the periodic function $E_k(\tau)$; equivalently, the Laurent expansion in $z = e^{2\pi i\tau}$.

Lemma 2.4

If $k \geq 2$ and $\mathrm{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell \tau}.$$

Proof. One may apply Poisson summation to $f(z) = 1/(z+\tau)^k$. Alternatively, it suffices to treat $k = 2$, the other cases following by termwise differentiation. Differentiating $\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau$ (Chapter 5) gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

With $w = e^{2\pi i\tau}$ ($|w| < 1$), $\sin^2 \pi\tau = -\frac{(e^{i\pi\tau} - e^{-i\pi\tau})^2}{4} = -\frac{(1-w)^2}{4w}$, so $\frac{\pi^2}{\sin^2 \pi\tau} = -4\pi^2 \frac{w}{(1-w)^2} = (-2\pi i)^2 \sum_{r=1}^{\infty} r w^r$, using $\sum_{r \geq 1} r w^r = \frac{w}{(1-w)^2}$. ∎

The **divisor function** $\sigma_\ell(r) = \sum_{d|r} d^\ell$ is the sum of the ℓth powers of the divisors of r.

Theorem 2.5

If $k \geq 4$ is even, and $\mathrm{Im}(\tau) > 0$, then

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. $\sigma_{k-1}(r) \leq \sum_{d \leq r} d^{k-1} \leq r^k$; if $\mathrm{Im}(\tau) \geq t_0 > 0$, $|e^{2\pi i r\tau}| \leq e^{-2\pi r t_0}$, so the series converges absolutely, by comparison with $\sum r^k e^{-2\pi r t_0}$. Using the definition, k even, and the lemma with τ replaced by $m\tau$,

$$\begin{aligned} E_k(\tau) &= \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} = 2\zeta(k) + 2 \sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2\frac{(-2\pi i)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m \tau} = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i r \tau}, \end{aligned}$$

collecting the terms with $\ell m = r$. ∎

For the "forbidden" case $k=2$, $\sum_{(n,m)\neq(0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but define

$$F(\tau)=\sum_m\sum_n \frac{1}{(n+m\tau)^2}, \qquad (n,m)\neq(0,0),$$

summed in the indicated order (first n, then m).

Corollary 2.6

The double sum defining F converges in the indicated order. We have

$$F(\tau)=2\zeta(2)-8\pi^2\sum_{r=1}^{\infty}\sigma(r)e^{2\pi i r\tau},$$

where $\sigma(r)=\sum_{d\mid r} d$ is the sum of the divisors of r.

(The argument of Theorem 2.5 applies with $k=2$.) $F(-1/\tau)\tau^{-2}\neq F(\tau)$; equivalently, summing first in m and then in n gives a different value. Nevertheless this forbidden Eisenstein series plays a crucial role in the four-squares theorem (Chapter 10).

3 Selected exercises

Motivation. These exercises from Stein–Shakarchi (numbers kept) justify the normalization of the periods, add Abel's relation for zeros and poles, show why the exponent $r>2$ in Lemma 1.5 is sharp, and evaluate $\zeta(2)$, $\zeta(4)$ from the lattice sums.

Exercise 9.1 (Real ratio of periods)

Let f be meromorphic on $\mathbb{C}$ with non-zero periods ω_1,ω_2 such that $\omega_2/\omega_1\in\mathbb{R}$.
(a) If $\omega_2/\omega_1=p/q$ with p,q relatively prime integers, $q\geq 1$, show that f has the periods ω_1,ω_2 if and only if f has the single period $\omega_0=\omega_1/q$.
(b) If ω_2/ω_1 is irrational, show that f is constant.

Solution. If ω is a period, so is $k\omega$ for every $k\in\mathbb{Z}$ (induction on $k\geq 0$, and $f(z-\omega)=f((z-\omega)+\omega)$); if ω,ω' are periods, so is $\omega+\omega'$. Hence all integer combinations of periods are periods.

(a)

Step 1. If ω_0 is a period, then $\omega_1=q\omega_0$ and $\omega_2=\frac{p}{q}\omega_1=p\omega_0$ are integer multiples of it, hence periods.

Step 2. Conversely, since $\gcd(p,q)=1$ there are integers m,n with $mq+np=1$. Then

$$m\omega_1+n\omega_2=\omega_1\Big(m+n\frac{p}{q}\Big)=\frac{\omega_1}{q}(mq+np)=\omega_0,$$

an integer combination of ω_1,ω_2, hence a period.

(b) Let $\alpha=\omega_2/\omega_1\notin\mathbb{Q}$ and $D=\{m+n\alpha : m,n\in\mathbb{Z}\}$.

Step 1. (D is dense in $\mathbb{R}$.) Fix $N\geq 1$. Among the $N+1$ fractional parts of $0,\alpha,\dots,N\alpha$, two, say those of $j\alpha$ and $j'\alpha$ with $j\neq j'$, lie in the same interval $[i/N,(i+1)/N)$. Hence $\delta=(j-j')\alpha-\ell$ for a suitable $\ell\in\mathbb{Z}$ satisfies $|\delta|<1/N$, and $\delta\neq 0$ because α is irrational. All $k\delta$, $k\in\mathbb{Z}$, lie in D and are spaced by $|\delta|<1/N$, so every interval of length $1/N$ meets D. As N is arbitrary, D is dense.

Step 2. (Many small periods.) For $t=m+n\alpha\in D$, $t\omega_1=m\omega_1+n\omega_2$ is a period. By Step 1 choose $t_k\in D\setminus\{0\}$ with $t_k\to 0$.

Step 3. (Conclusion.) The set S of poles of f is discrete and closed, so $U = \mathbb{C} \setminus S$ is open and connected. Fix $z_0 \in U$. The points $z_k = z_0 + t_k\omega_1$ satisfy $z_k \neq z_0$, $z_k \to z_0$ and $f(z_k) = f(z_0)$ by Step 2. Thus the holomorphic function $f - f(z_0)$ on U has zeros accumulating at $z_0 \in U$, so $f \equiv f(z_0)$ on U. A function constant on a punctured neighborhood of a point of S cannot have a pole there, so $S = \varnothing$ and f is constant. ∎

Exercise 9.2 (Zeros minus poles)

Let f be an elliptic function with periods $1, \tau$ and lattice Λ, and let $a_1, \dots, a_r$ and $b_1, \dots, b_r$ be its zeros and poles in P_0, repeated according to multiplicity (there are equally many by Theorem 1.4). Show that

$$a_1 + \cdots + a_r - b_1 - \cdots - b_r \in \Lambda.$$

Solution.

Step 1. (Choice of a parallelogram.) The set Z of zeros and poles of f is countable. For $w \in Z$, the set of h with $w \in \partial(h + P_0)$ is $w - \partial P_0$, a union of four segments, which has area zero. A countable union of such sets has area zero, so there is h with no zero or pole on ∂P, where $P = h + P_0$.

Step 2. (Reduction to P.) By Proposition 1.1, each $z \in P_0$ is congruent to a unique $\phi(z) \in P$, and $\phi : P_0 \to P$ is a bijection with $\phi(z) - z \in \Lambda$. Since $f(z + \lambda) = f(z)$ for $\lambda \in \Lambda$, f has a zero (pole) of order k at z if and only if it has one at $\phi(z)$. Hence the zeros and poles in P are $a_i' = \phi(a_i)$ and $b_j' = \phi(b_j)$, and

$$\textstyle\sum_i a_i - \sum_j b_j = \sum_i a_i' - \sum_j b_j' - \lambda \qquad \text{for some } \lambda \in \Lambda.$$

So it suffices to show $\sum_i a_i' - \sum_j b_j' \in \Lambda$.

Step 3. (Residues.) Near a zero a of order k, $f'/f = k/(z-a) + (\text{holomorphic})$, so zf'/f has residue ka there; at a pole b of order k the residue is $-kb$. Elsewhere near $\overline{P}$ the function zf'/f is holomorphic. By the residue theorem,

$$\frac{1}{2\pi i}\int_{\partial P} z\frac{f'(z)}{f(z)}\,dz = \sum_i a_i' - \sum_j b_j'.$$

Step 4. (Pairing opposite sides.) Let $g = f'/f$, which has periods 1 and τ. The boundary runs $h \to h+1 \to h+1+\tau \to h+\tau \to h$. Substituting $z \mapsto z+\tau$ in the third side and $z \mapsto z+1$ in the second,

$$\int_h^{h+1} zg\,dz + \int_{h+1+\tau}^{h+\tau} zg\,dz = \int_h^{h+1} [z - (z+\tau)]g(z)\,dz = -\tau\int_h^{h+1} g(z)\,dz,$$
$$\int_{h+1}^{h+1+\tau} zg\,dz + \int_{h+\tau}^{h} zg\,dz = \int_h^{h+\tau} [(z+1) - z]g(z)\,dz = \int_h^{h+\tau} g(z)\,dz.$$

Step 5. (Integers.) The curve $\gamma(t) = f(h+t)$, $0 \le t \le 1$, is closed because $f(h+1) = f(h)$, and avoids 0. Hence

$$\int_h^{h+1} \frac{f'(z)}{f(z)}\,dz = \int_\gamma \frac{dw}{w} = 2\pi i\, n$$

with $n \in \mathbb{Z}$ the winding number of γ about 0. In the same way $\int_h^{h+\tau} g = 2\pi i\,\ell$ with $\ell \in \mathbb{Z}$.

Step 6. Combining Steps 3–5, $\sum_i a_i' - \sum_j b_j' = \ell - n\tau \in \Lambda$. ∎

Exercise 9.3 (The exponent 2 is critical)

(a) Show that $\sum_{\omega\in\Lambda^*}|\omega|^{-2}$ diverges for every $\tau\in\mathbb{H}$.
(b) Show that, as $R\to\infty$,

$$\sum_{1\le n^2+m^2\le R^2}\frac{1}{n^2+m^2}=2\pi\log R+O(1).$$

Solution.

(b) Write $\omega=(n,m)\in\mathbb{Z}^2$, $f(x)=|x|^{-2}$ on $\mathbb{R}^2\setminus\{0\}$, $c=\sqrt{2}/2$, and let Q_ω be the closed unit square centered at ω. Take $R\ge 4$.

Step 1. (Local comparison.) Let $|\omega|\ge 2$ and $x\in Q_\omega$. Every y on the segment $[\omega,x]\subset Q_\omega$ has $|y|\ge|\omega|-c\ge|\omega|/2$, and $|\nabla f(y)|=2|y|^{-3}\le 16|\omega|^{-3}$. The segment has length at most c, so by the mean value theorem

$$|f(x)-f(\omega)|\le 8\sqrt{2}\,|\omega|^{-3}.$$

Step 2. (Sum versus integral.) Split the sum into the six terms with $1\le|\omega|<2$, namely $(\pm1,0)$, $(0,\pm1)$, $(\pm1,\pm1)$, whose total is $4+4\cdot\frac12=6$, and the sum $S_1(R)$ over $2\le|\omega|\le R$. The squares have area 1 and disjoint interiors; let U_R be the union of Q_ω, $2\le|\omega|\le R$. By Step 1,

$$\Big|S_1(R)-\int_{U_R}f\,dx\Big|\le 8\sqrt{2}\sum_{|\omega|\ge2}|\omega|^{-3}=:C<\infty,$$

the series converging by Lemma 1.5 with $r=3$, $\tau=i$.

Step 3. (Annuli.) Let $A(a,b)=\{a\le|x|\le b\}$. If $2+c\le|x|\le R-c$, then x lies in some Q_ω, and $|x-\omega|\le c$ gives $2\le|\omega|\le R$; so $A(2+c,R-c)\subset U_R$. Conversely, $x\in Q_\omega$ with $2\le|\omega|\le R$ has $2-c\le|x|\le R+c$, so $U_R\subset A(2-c,R+c)$. In polar coordinates $\int_{A(a,b)}f=\int_0^{2\pi}\int_a^b r^{-1}\,dr\,d\theta=2\pi\log(b/a)$, and $f>0$, so

$$2\pi\log\frac{R-c}{2+c}\le\int_{U_R}f\,dx\le 2\pi\log\frac{R+c}{2-c}.$$

Step 4. (Conclusion.) For $R\ge4$, $|\log(1\pm c/R)|\le\log 2$, so both bounds equal $2\pi\log R+O(1)$. With Step 2 the full sum is $6+2\pi\log R+O(1)=2\pi\log R+O(1)$.

(a) With $\tau\in\mathbb{H}$ and $\omega=n+m\tau\neq0$,

$$|n+m\tau|\le|n|+|m||\tau|\le(1+|\tau|)(|n|+|m|)\le\sqrt{2}(1+|\tau|)(n^2+m^2)^{1/2}.$$

Hence $|\omega|^{-2}\ge\frac{1}{2(1+|\tau|)^2}(n^2+m^2)^{-1}$, and the partial sums over $n^2+m^2\le R^2$ are at least a fixed multiple of $2\pi\log R+O(1)\to\infty$ by (b). So the series diverges. ∎

Exercise 9.7 ($\zeta(2)$ and $\zeta(4)$ from lattice sums)

Using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-2}=\pi^2/\sin^2(\pi\tau)$ at $\tau=1/2$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^2}=\frac{\pi^2}{8},\qquad\zeta(2)=\frac{\pi^2}{6},$$

and, using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-4}$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^4}=\frac{\pi^4}{96},\qquad\zeta(4)=\frac{\pi^4}{90}.$$

Solution.

Step 1. (The identities on $\mathbb{C}\setminus\mathbb{Z}$.) By Chapter 5, $\pi\cot\pi z = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ for $z\notin\mathbb{Z}$, uniformly on compact subsets of $\mathbb{C}\setminus\mathbb{Z}$. Differentiating termwise (Weierstrass), as in the proof of Lemma 2.4,

$$\sum_{n\in\mathbb{Z}} \frac{1}{(z+n)^2} = \frac{\pi^2}{\sin^2\pi z} =: g(z), \qquad z\in\mathbb{C}\setminus\mathbb{Z},$$

and the left side converges absolutely and locally uniformly, so it may be differentiated twice more termwise:

$$\sum_{n\in\mathbb{Z}} \frac{6}{(z+n)^4} = g''(z).$$

Step 2. (Rewriting at $z=1/2$.) $(n+\frac12)^{-k} = 2^k(2n+1)^{-k}$, and as n runs over $\mathbb{Z}$, $2n+1$ runs over the odd integers, each odd $m\ge 1$ arising twice ($\pm m$). So for even k,

$$\sum_{n\in\mathbb{Z}} \frac{1}{(n+\frac12)^k} = 2^{k+1} \sum_{m\ge 1 \text{ odd}} \frac{1}{m^k}.$$

Step 3. ($k=2$.) $g(1/2)=\pi^2$, so $\sum_{m\text{ odd}} m^{-2} = \pi^2/8$. Splitting $\zeta(2)$ into odd and even terms, $\zeta(2) = \frac{\pi^2}{8} + \frac14\zeta(2)$, hence $\zeta(2)=\pi^2/6$.

Step 4. ($k=4$.) With $u=\pi z$,

$$g'(z) = -2\pi^3\csc^2 u\cot u, \qquad g''(z) = 2\pi^4\big(2\csc^2 u\cot^2 u + \csc^4 u\big).$$

At $z=1/2$, $\csc u = 1$ and $\cot u = 0$, so $g''(1/2) = 2\pi^4$ and $\sum_n (n+\frac12)^{-4} = \pi^4/3$. By Step 2, $\sum_{m\text{ odd}} m^{-4} = \frac{\pi^4}{3\cdot 32} = \frac{\pi^4}{96}$. Then $\zeta(4) = \frac{\pi^4}{96} + \frac{1}{16}\zeta(4)$, so $\zeta(4) = \frac{16}{15}\cdot\frac{\pi^4}{96} = \frac{\pi^4}{90}$. ∎

Chapter 10. Applications of Theta Functions

Overview

The *theta function*

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

converges for all $z \in \mathbb{C}$ and τ in the upper half-plane. It has a dual nature: as a function of z it belongs to the world of elliptic functions (period 1, "quasi-period" τ); as a function of τ it is modular and connects with partitions and with representing integers as sums of squares. The main tools are the *triple product formula* and the *transformation law*. We then discuss partitions and prove the celebrated two-squares and four-squares theorems.

1 Product formula for the Jacobi theta function

Motivation. The theta function $\Theta(z|\tau)$ is an entire function with explicit zeros, and its Jacobi product formula is the source of many identities in number theory, including the generating functions and the sums of squares theorems at the end of the chapter.

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, **Jacobi's theta function** is

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}. \tag{1}$$

Special cases: $\theta(\tau) = \sum_n e^{\pi i n^2 \tau} = \Theta(0|\tau)$ for $\tau \in \mathbb{H}$, and $\vartheta(t) = \sum_n e^{-\pi n^2 t} = \theta(it)$ for $t > 0$. The heat kernel on the circle (Book I) is $H_t(x) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = \Theta(x|4\pi i t)$; and the functional equation of ϑ gave that of ζ (Chapter 6).

Proposition 1.1

The function Θ satisfies the following properties:
(i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
(ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
(iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$.
(iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. If $\operatorname{Im}(\tau) = t \geq t_0 > 0$ and $|z| \leq M$, then $\sum_n |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty$, uniformly; this gives (i).

(ii) is clear since $e^{2\pi i n z}$ has period 1. For (iii), complete the square:

$$\Theta(z+\tau|\tau) = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} = \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1) z} e^{-2\pi i z} = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}.$$

By (ii), (iii) it suffices for (iv) to show $\Theta(1/2 + \tau/2|\tau) = 0$: $\Theta(1/2+\tau/2|\tau) = \sum_n (-1)^n e^{\pi i (n^2+n)\tau}$, and pairing $n \geq 0$ with $-n-1$, which has opposite parity and $(-n-1)^2 + (-n-1) = n^2 + n$, the terms cancel. ∎

With the standard notation $q = e^{\pi i \tau}$, the **triple product** is

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 1.2

The function $\Pi(z|\tau)$ satisfies the following properties:
(i) $\Pi(z|\tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
(ii) $\Pi(z+1|\tau) = \Pi(z|\tau)$.
(iii) $\Pi(z+\tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
(iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\mathrm{Im}(\tau) \geq t_0 > 0$, $|q| \leq e^{-\pi t_0} < 1$ and each factor is $1+O(|q|^{2n-1}e^{2\pi|z|})$; since $\sum|q|^{2n-1} < \infty$, the results on infinite products (Section 3 of Chapter 5) give (i).

(ii) is clear. For (iii), since $q^2 = e^{2\pi i\tau}$,

$$\Pi(z+\tau|\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n+1}e^{2\pi iz})(1+q^{2n-3}e^{-2\pi iz}),$$

and comparing with $\Pi(z|\tau)$, $\Pi(z+\tau|\tau) = \Pi(z|\tau)\frac{1+q^{-1}e^{-2\pi iz}}{1+qe^{2\pi iz}}$;

(iii) follows from $(1+x)/(1+x^{-1}) = x$ for $x \neq -1$. For (iv), a convergent product vanishes only if a factor does. $1-q^{2n} \neq 0$. $1+q^{2n-1}e^{2\pi iz} = 0$ iff $q^{2n-1}e^{2\pi iz} = -1 = e^{\pi i}$, i.e. $(2n-1)\tau + 2z \equiv 1 \pmod 2$, i.e. $z \equiv 1/2 + \tau/2 - n\tau \pmod 1$ $(n \geq 1)$. $1+q^{2n-1}e^{-2\pi iz} = 0$ iff $(2n-1)\tau - 2z \equiv 1 \pmod 2$, i.e. $z \equiv -1/2 - \tau/2 + n\tau \equiv 1/2 + \tau/2 + (n-1)\tau \pmod 1$ $(n \geq 1)$. This exhausts the zeros, and they are simple since $e^w - 1$ vanishes to order 1 at 0. ∎

Theorem 1.3 (Product formula)

For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.

Proof. Fix $\tau \in \mathbb{H}$. There is a constant $c(\tau)$ with

$$\Theta(z|\tau) = c(\tau)\Pi(z|\tau)\,: \tag{2}$$

$F(z) = \Theta(z|\tau)/\Pi(z|\tau)$ is entire (by (iv) of both propositions) and doubly periodic with periods 1, τ (by (ii), (iii)), hence constant. We show $c(\tau) = c(4\tau)$. Put $z = 1/2$ in (2) ($e^{\pm 2\pi iz} = -1$):

$$\sum_n (-1)^n q^{n^2} = c(\tau)\prod_{n=1}^{\infty}(1-q^{2n})(1-q^{2n-1})^2 = c(\tau)\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1}), \quad c(\tau) = \frac{\sum_n (-1)^n q^{n^2}}{\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1})}. \tag{3}$$

Put $z = 1/4$ ($e^{2\pi iz} = i$): $\Theta(1/4|\tau) = \sum_n q^{n^2} i^n$, and since $1/i = -i$, only even $n = 2m$ survive: $\Theta(1/4|\tau) = \sum_m q^{4m^2}(-1)^m$. On the other hand

$$\Pi(1/4|\tau) = \prod_{m=1}^{\infty}(1-q^{2m})(1+iq^{2m-1})(1-iq^{2m-1}) = \prod_{m=1}^{\infty}(1-q^{2m})(1+q^{4m-2}) = \prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4}),$$

the last step by splitting $2m = 4n$ and $2m = 4n-2$ in the first factor ($(1-q^{4n-2})(1+q^{4n-2}) = 1-q^{8n-4}$). Hence

$$c(\tau) = \frac{\sum_n (-1)^n q^{4n^2}}{\prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4})}, \tag{4}$$

and comparing (3) and (4) ($q \mapsto q^4$ corresponds to $\tau \mapsto 4\tau$), $c(\tau) = c(4\tau)$. So $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{\pi i 4^k \tau} \to 0$, (3) gives $c(4^k\tau) \to 1$; so $c(\tau) = 1$. ∎

Corollary 1.4

If $\operatorname{Im}(\tau) > 0$ and $q = e^{\pi i \tau}$, then

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

Corollary 1.5

For each fixed $\tau \in \mathbb{H}$, the quotient

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ, and with a double pole at $z = 1/2 + \tau/2$.

(Primes denote differentiation in z.)

Proof. Let $F = (\log \Theta)' = \Theta'/\Theta$. Differentiating (ii), (iii) of Proposition 1.1 gives $F(z+1) = F(z)$ and $F(z+\tau) = F(z) - 2\pi i$, so F' is doubly periodic. Θ vanishes in the fundamental parallelogram only at $1/2 + \tau/2$, simply, so F has only a simple pole there and F' only a double pole. ∎

1.1 Further transformation laws

In Chapter 9 the modular character was expressed through $T_1 : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. For Θ it is natural to use $T_2 : \tau \mapsto \tau + 2$ and S, since $\Theta(z|\tau + 2) = \Theta(z|\tau)$ but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Theorem 1.6

If $\tau \in \mathbb{H}$, then

$$\Theta(z|-1/\tau) = \sqrt{\tau/i}\, e^{\pi i z^2 \tau} \Theta(z\tau|\tau) \qquad \text{for all } z \in \mathbb{C}. \tag{5}$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to prove (5) for $z = x$ real and $\tau = it$, $t > 0$: for fixed real x both sides are holomorphic in $\tau \in \mathbb{H}$ and would agree on the positive imaginary axis; and for fixed τ both sides are entire in z and would agree on $\mathbb{R}$. For $z = x$, $\tau = it$, (5) reads

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a, this is $\sum_n e^{-\pi t(n+a)^2} = \sum_n t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}$, which is equation (3) of Chapter 4 (from Poisson summation). ∎

Corollary 1.7

If $\operatorname{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i}\, \theta(\tau)$.

(Set $z = 0$; for $\tau = it$ this is the functional equation of ϑ.) This gives precise information about θ as $\tau \to 0$; the next corollary describes θ near $\tau = 1$.

Corollary 1.8

If $\tau \in \mathbb{H}$, then

$$\theta(1-1/\tau) = \sqrt{\tau/i}\sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2\tau} = \sqrt{\tau/i}\,(2e^{\pi i\tau/4}+\cdots).$$

The second identity means that $\theta(1-1/\tau) \sim 2\sqrt{\tau/i}\,e^{\pi i\tau/4}$ as $\mathrm{Im}(\tau)\to\infty$.

Proof. n and n^2 have the same parity, so $\theta(1+\tau) = \sum_n (-1)^n e^{\pi i n^2\tau} = \Theta(1/2|\tau)$; hence $\theta(1-1/\tau) = \Theta(1/2|-1/\tau)$. By Theorem 1.6 with $z = 1/2$,

$$\theta(1-1/\tau) = \sqrt{\tau/i}\,e^{\pi i\tau/4}\Theta(\tau/2|\tau) = \sqrt{\tau/i}\,e^{\pi i\tau/4}\sum_n e^{\pi i n^2\tau}e^{\pi i n\tau} = \sqrt{\tau/i}\sum_n e^{\pi i(n+1/2)^2\tau}.$$

The terms $n = 0, -1$ contribute $2e^{\pi i\tau/4}$, of absolute value $2e^{-\pi t/4}$ ($\tau = \sigma + it$); the others total $O(\sum_{k\ge1} e^{-\pi(k+1/2)^2 t}) = O(e^{-9\pi t/4})$. ∎

The **Dedekind eta function** is, for $\mathrm{Im}(\tau) > 0$,

$$\eta(\tau) = e^{\frac{\pi i\tau}{12}}\prod_{n=1}^{\infty}(1-e^{2\pi i n\tau}).$$

Proposition 1.9

If $\mathrm{Im}(\tau) > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$.

Proof. Differentiate (5) and evaluate at $z_0 = 1/2 + \tau/2$. By the product formula,

$$\Theta(z|\tau) = (1+qe^{-2\pi iz})\prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1}e^{2\pi iz})(1+q^{2n+1}e^{-2\pi iz}),$$

and the first factor vanishes at z_0 (where $e^{2\pi iz_0} = -q$), with derivative $-2\pi iqe^{-2\pi iz_0} = 2\pi i$ there, while the remaining factors equal $(1-q^{2n})^3$. So

$$\Theta'(z_0|\tau) = 2\pi i H(\tau), \qquad H(\tau) = \prod_{n=1}^{\infty}(1-e^{2\pi i n\tau})^3.$$

Replacing $-1/\tau$ by τ in (5), $\Theta(z|\tau) = \sqrt{i/\tau}\,e^{-\pi iz^2/\tau}\Theta(-z/\tau|-1/\tau)$. Differentiating and evaluating at z_0 (where $-z_0/\tau = -\frac{1}{2\tau} - \frac{1}{2}$ is a zero of $\Theta(\cdot|-1/\tau)$, congruent to $\frac{1}{2} + \frac{-1/\tau}{2}$), a computation gives

$$2\pi i H(\tau) = \sqrt{i/\tau}\,e^{-\frac{\pi i\tau}{4}}e^{-\frac{\pi i}{2}}e^{-\frac{\pi i}{4\tau}}\Big(-\frac{1}{\tau}\Big)2\pi i H(-1/\tau), \qquad \text{i.e.} \qquad e^{\frac{\pi i\tau}{4}}H(\tau) = \Big(\frac{i}{\tau}\Big)^{3/2}e^{-\frac{\pi i}{4\tau}}H(-1/\tau),$$

which says $\eta(\tau)^3 = (i/\tau)^{3/2}\eta(-1/\tau)^3$. For $\tau = it$, $\eta(\tau) > 0$, so taking cube roots gives $\eta(\tau) = \sqrt{i/\tau}\,\eta(-1/\tau)$; by analytic continuation this holds on $\mathbb{H}$. ∎

2 Generating functions

For a sequence $\{F_n\}_{n=0}^{\infty}$, its **generating function** is $F(x) = \sum_{n=0}^{\infty} F_n x^n$; properties of the sequence translate into algebraic or analytic properties of F, which can lead back to new insights. The **partition function** $p(n)$ is the number of ways n can be written as a sum of positive integers ($p(0) = 1$; e.g. $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$, $p(6) = 11$, $p(7) = 15$, $p(8) = 22$, $p(12) = 77$).

Theorem 2.1 (Euler)

If $|x| < 1$, then $\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.

Formally, write $\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km}$ and multiply out: grouping equal parts, a partition is $n = m_1k_1 + \cdots + m_rk_r$ with distinct k_j, corresponding to the term $(x^{k_1})^{m_1} \cdots (x^{k_r})^{m_r}$. The justification is as for the product formula for ζ (Section 1 of Chapter 7), using convergence of $\prod 1/(1-x^k)$, since $\frac{1}{1-x^k} = 1 + O(x^k)$.

Similarly $\prod 1/(1-x^{2n-1})$ generates $p_o(n)$, the number of partitions of n into odd parts, and $\prod(1+x^n)$ generates $p_u(n)$, partitions into unequal parts. Remarkably $p_o(n) = p_u(n)$, i.e.

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty} (1+x^n) :$$

indeed $\prod(1+x^n)\prod(1-x^n) = \prod(1-x^{2n})$, and by parity $\prod(1-x^{2n})\prod(1-x^{2n-1}) = \prod(1-x^n)$.

Let $p_{e,u}(n)$, $p_{o,u}(n)$ be the numbers of partitions of n into an even, resp. odd, number of unequal parts. Euler proved $p_{e,u}(n) = p_{o,u}(n)$ unless n is a **pentagonal number** $k(3k+1)/2$, $k \in \mathbb{Z}$ (namely $1, 2, 5, 7, 12, 15, 22, 26, \ldots$), in which case $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$. Since multiplying out gives terms $(-1)^r x^{n_1+\cdots+n_r}$ with distinct n_j,

$$\prod_{n=1}^{\infty} (1-x^n) = \sum_{n=0}^{\infty} [p_{e,u}(n) - p_{o,u}(n)]x^n,$$

so Euler's theorem follows from:

Proposition 2.2 (Euler's pentagonal number theorem)

$$\prod_{n=1}^{\infty} (1-x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. Set $x = e^{2\pi iu}$. In the triple product take $q = e^{3\pi iu}$ and $z = 1/2 + u/2$; then

$$\prod_{n=1}^{\infty} (1 - e^{2\pi i 3nu})(1 - e^{2\pi i(3n-1)u})(1 - e^{2\pi i(3n-2)u}) = \prod_{n=1}^{\infty} (1 - e^{2\pi inu}).$$

By Theorem 1.3 the product equals $\sum_n e^{3\pi in^2u}(-1)^n e^{2\pi inu/2} = \sum_n (-1)^n e^{\pi in(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}$. ∎

The growth of $p(n)$ can be analyzed via $1/\prod(1-x^n)$ as $|x| \to 1$; elementary considerations give a rough order of growth, and the transformation law of η (Proposition 1.9) leads to a precise asymptotic formula (Appendix A).

3 The theorems about sums of squares

Let $r_2(n)$ be the number of pairs $(x, y) \in \mathbb{Z}^2$ with $n = x^2 + y^2$ (e.g. $r_2(3) = 0$, $r_2(5) = 8$). *Sum of two squares:* which integers are sums of two squares, and what is $r_2(n)$? No integer $4k+3$ is a sum of two squares, and no integer $8k+7$ is a sum of three squares. Let $r_4(n)$ be the number of representations of n as a sum of four squares. *Sum of four squares:* is every positive integer a sum of four squares, and what is $r_4(n)$? These problems, going back to the third century, were first fully solved with Jacobi's theta functions.

3.1 The two-squares theorem

If $n = a^2 + b^2$ and $m = c^2 + d^2$, then with $x + iy = (a + ib)(c + id)$, $x^2 + y^2 = nm$; so divisibility properties of n matter. Let $d_1(n)$ and $d_3(n)$ be the numbers of divisors of n of the form $4k+1$ and $4k+3$.

Theorem 3.1 (Two-squares theorem)

If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.

Consequently, if $n = p_1^{a_1} \cdots p_r^{a_r}$ with distinct primes p_j, then n is a sum of two squares iff every prime p_j of the form $4k+3$ has an even exponent a_j. The starting point: for $q = e^{\pi i \tau}$, $\tau \in \mathbb{H}$,

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n, \tag{6}$$

since $\theta(\tau)^2 = \sum_{n_1} \sum_{n_2} q^{n_1^2} q^{n_2^2} = \sum_{(n_1,n_2)\in\mathbb{Z}^2} q^{n_1^2+n_2^2}$.

Proposition 3.2

The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities

$$\theta(\tau)^2 = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1+q^{2n}}, \tag{7}$$

whenever $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$.

Proof. Both series converge absolutely ($|q| < 1$), and they are equal since $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$. Since $(1+q^{2n})^{-1} = (1-q^{2n})/(1-q^{4n})$, the right side of (7) is $1 + 4\sum_{n\geq 1}\left[\frac{q^n}{1-q^{4n}} - \frac{q^{3n}}{1-q^{4n}}\right]$. Now $\frac{1}{1-q^{4n}} = \sum_{m\geq 0} q^{4nm}$, so

$$\sum_{n=1}^{\infty} \frac{q^n}{1-q^{4n}} = \sum_{n=1}^{\infty}\sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k) q^k, \qquad \sum_{n=1}^{\infty} \frac{q^{3n}}{1-q^{4n}} = \sum_{k=1}^{\infty} d_3(k) q^k$$

(the series converge since $d_j(k) \leq k$). Compare with (6). ∎

Let

$$C(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(\pi n \tau)}, \qquad q = e^{\pi i \tau},\ \tau \in \mathbb{H}. \tag{8}$$

We must prove $\theta^2 = C$. There is a striking parallel: θ comes from the heat equation on the line, whose kernel is the Gaussian $e^{-\pi x^2}$, its own Fourier transform, and its transformation law comes from Poisson summation; C comes from the steady-state heat equation in a strip, whose kernel is $1/\cosh \pi x$ (Section 1), again its own Fourier transform (Example 3 in Section 2 of Chapter 3), and its transformation law again comes from Poisson summation. Indeed, $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7); and (5) of Chapter 4 with $a = 0$ gives $\sum_n \frac{1}{\cosh \pi n t} = \frac{1}{t}\sum_n \frac{1}{\cosh(\pi n/t)}$, i.e. $C(\tau) = (i/\tau)C(-1/\tau)$ for $\tau = it$, hence for all $\tau \in \mathbb{H}$. Both θ^2 and C tend to 1 as $\mathrm{Im}(\tau) \to \infty$. At the "cusp" $\tau = 1$: by Corollary 1.8, $\theta(1-1/\tau)^2 \sim 4(\tau/i)e^{\pi i \tau/2}$; and (5) of Chapter 4 with $a = 1/2$ gives $\sum_n \frac{(-1)^n}{\cosh(\pi n/t)} = t\sum_n \frac{1}{\cosh(\pi(n+1/2)t)}$, so by analytic continuation

$$C(1-1/\tau) = \frac{\tau}{i} \sum_n \frac{1}{\cos(\pi(n+1/2)\tau)} = 4\frac{\tau}{i} e^{\pi i \tau/2} + O(|\tau| e^{-3\pi t/2}) \quad (t \to \infty),$$

the main terms being $n = 0, -1$.

Proposition 3.3

The function $C(\tau) = \sum 1/\cos(\pi n\tau)$, defined in the upper half-plane, satisfies (i) $C(\tau+2) = C(\tau)$.
(ii) $C(\tau) = (i/\tau)C(-1/\tau)$.
(iii) $C(\tau) \to 1$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $C(1-1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\mathrm{Im}(\tau) \to \infty$. Moreover, $\theta(\tau)^2$ satisfies the same properties.

Theorem 3.4

Suppose f is a holomorphic function in the upper half-plane that satisfies:
(i) $f(\tau+2) = f(\tau)$, (ii) $f(-1/\tau) = f(\tau)$, (iii) $f(\tau)$ is bounded, then f is constant.

Let $\mathcal{F} = \{\tau \in \overline{\mathbb{H}} : |\mathrm{Re}(\tau)| \le 1,\ |\tau| \ge 1\}$; the points $\tau = \pm 1$ are its **cusps** (equivalent under $\tau \mapsto \tau+2$).

Lemma 3.5

Every point in the upper half-plane can be mapped into $\mathcal{F}$ using repeatedly one or another of the following fractional linear transformations or their inverses: $T_2 : \tau \mapsto \tau+2$, $S : \tau \mapsto -1/\tau$.

$\mathcal{F}$ is the **fundamental domain** for the group G generated by T_2 and S (uniqueness of representatives fails only on $\partial\mathcal{F}$). Elements $g \in G$ are represented by matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $g(\tau) = \frac{a\tau+b}{c\tau+d}$; these have integer entries and determinant 1, so

$$\mathrm{Im}(g(\tau)) = \frac{\mathrm{Im}(\tau)}{|c\tau+d|^2}. \tag{9}$$

Proof. For $\tau \in \mathbb{H}$, since c, d are integers, by (9) we may choose $g_0 \in G$ with $\mathrm{Im}(g_0(\tau))$ maximal. Translations $T_2^{\pm 1}$ do not change imaginary parts, so there is $g_1 \in G$ with $|\mathrm{Re}(g_1(\tau))| \le 1$ and $\mathrm{Im}(g_1(\tau))$ maximal. If $|g_1(\tau)| < 1$, then $\mathrm{Im}(Sg_1(\tau)) = \mathrm{Im}(-1/g_1(\tau)) = \mathrm{Im}(g_1(\tau))/|g_1(\tau)|^2 > \mathrm{Im}(g_1(\tau))$, contradicting maximality. So $g_1(\tau) \in \mathcal{F}$. ∎

Proof of Theorem 3.4. Suppose f is not constant. Let $g(z) = f(\tau)$ with $z = e^{\pi i\tau}$; g is well defined in the punctured unit disc (period 2) and bounded near 0, so 0 is removable and $\lim_{\mathrm{Im}\,\tau\to\infty} f(\tau)$ exists; by the maximum modulus principle $\lim_{\mathrm{Im}\,\tau\to\infty}|f(\tau)| < \sup_{\mathcal{F}}|f|$. At the cusps (by $f(\tau+2) = f(\tau)$ it suffices to treat $\tau = 1$) we show $\lim_{\mathrm{Im}\,\tau\to\infty} f(1-1/\tau)$ exists and $\lim|f(1-1/\tau)| < \sup_{\mathcal{F}}|f|$, by interchanging 1 and ∞. Let U_n be the fractional linear transformation of $\begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$, which fixes 1; with $\nu(\tau) = 1/(1-\tau)$ (taking 1 to ∞, with inverse $\nu^{-1}(\tau) = 1-1/\tau$), $U_n = \nu^{-1}T_n\nu$, $T_n(\tau) = \tau+n$. Hence $U_nU_m = U_{n+m}$ and $U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = T_2S$. So every U_n is a finite composite of T_2, S and their inverses, and f is invariant under U_n: $f(\nu^{-1}T_n\nu(\tau)) = f(\tau)$. Thus $F(\tau) = f(\nu^{-1}(\tau)) = f(1-1/\tau)$ has period 1, and as above $h(z) = F(\tau)$, $z = e^{2\pi i\tau}$, has a removable singularity at 0, giving the claim by the maximum principle. Hence f attains its maximum in the interior of $\mathbb{H}$, contradicting the maximum principle. ∎

Proof of the two-squares theorem. $f(\tau) = C(\tau)/\theta(\tau)^2$ is holomorphic in $\mathbb{H}$ ($\theta \neq 0$ by Corollary 1.4) and invariant under T_2 and S (Proposition 3.3). In $\mathcal{F}$ it is bounded, tending to 1 as $\mathrm{Im}(\tau) \to \infty$ or $\tau \to \pm 1$, by (iii), (iv) of Proposition 3.3, verified by both C and θ^2. So f is bounded in $\mathbb{H}$, hence constant by Theorem 3.4, and the constant is 1: $\theta^2 = C$. ∎

3.2 The four-squares theorem

Let $\sigma_1^*(n)$ be the sum of the divisors of n that are not divisible by 4.

Theorem 3.6 (Four-squares theorem)

Every positive integer is the sum of four squares, and moreover $r_4(n) = 8\sigma_1^*(n)$ for all $n \geq 1$.

As before, $\theta(\tau)^4 = \sum_{n\geq 0} r_4(n)q^n$, $q = e^{\pi i\tau}$. Here nothing as simple as C is available; we need a subtle variant of the Eisenstein series. Recall the forbidden Eisenstein series $F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}$ (sum first in n, then m; $(n,m) \neq (0,0)$) of Corollary 2.6, and define

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau), \qquad \tau \in \mathbb{H}, \tag{10}$$

equivalently $E_2^*(\tau) = \sum_m \sum_n \left[\frac{1}{(n+m\tau/2)^2} - \frac{1}{(n/2+m\tau)^2}\right]$; the order of summation is critical.

Proposition 3.7

The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity $\theta(\tau)^4 = -\pi^{-2}E_2^*(\tau)$, where $\tau \in \mathbb{H}$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau) = 1 + 8\sum_{k\geq 1}\sigma_1^*(k)q^k$. By Corollary 2.6, $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_{k\geq 1}\sigma_1(k)e^{2\pi ik\tau}$, $\sigma_1(k)$ the sum of divisors of k. Now $\sigma_1^*(n) = \sigma_1(n)$ if $4 \nmid n$, and $\sigma_1^*(n) = \sigma_1(n) - 4\sigma_1(n/4)$ if $4 \mid n$ (a divisor $d = 4\tilde{d}$ of $n = 4\tilde{n}$ corresponds to the divisor $\tilde{d}$ of $\tilde{n}$). With (10),

$$E_2^*(\tau) = -\pi^2 - 8\pi^2\sum_{k=1}^{\infty}\sigma_1^*(k)e^{\pi ik\tau}.$$

■

So Theorem 3.6 reduces to $\theta^4 = -\pi^{-2}E_2^*$, and the key is that E_2^* has the same modular properties as θ^4.

Proposition 3.8

The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:
(i) $E_2^*(\tau+2) = E_2^*(\tau)$.
(ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$.
(iii) $E_2^*(\tau) \to -\pi^2$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $|E_2^*(1-1/\tau)| = O(|\tau^2 e^{\pi i\tau}|)$ as $\mathrm{Im}(\tau) \to \infty$. Moreover $-\pi^2\theta^4$ has the same properties.

(i) is immediate from the definition. Consider F and its *reverse* $\tilde{F}(\tau) = \sum_n \sum_m \frac{1}{(n+m\tau)^2}$ (summing first in m).

Lemma 3.9

The functions F and $\tilde{F}$ satisfy:
(a) $F(-1/\tau) = \tau^2\tilde{F}(\tau)$, (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$, (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$.

Proof. (a) follows from the definitions and $(n+m(-1/\tau))^2 = \tau^{-2}(-m+n\tau)^2$. For (b) use $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$ with $\eta(\tau) = q^{1/12}\prod(1-q^{2n})$, $q = e^{\pi i\tau}$. By Proposition 3.2 of Chapter 5, the logarithmic derivative in τ is

$$\frac{\eta'}{\eta}(\tau) = \frac{\pi i}{12} - 2\pi i\sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}}, \qquad \sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} nq^{2nm} = \sum_{k=1}^{\infty}\sigma_1(k)q^{2k}.$$

Since $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_k \sigma_1(k) q^{2k}$, $\frac{\eta'}{\eta}(\tau) = \frac{i}{4\pi} F(\tau)$. By the chain rule the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}\frac{\eta'}{\eta}(-1/\tau)$, which by (a) equals $\frac{i}{4\pi}\tilde{F}(\tau)$. Taking logarithmic derivatives in the functional equation for η,

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi}F(\tau),$$

so $\tilde{F}(\tau) = F(\tau) - 2\pi i/\tau$.

(c) follows from (a) and (b). ∎

Proof of Proposition 3.8. (ii) By (10) and Lemma 3.9(c),

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) = [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i\tau] \\ &= 4\tau^2 F(2\tau) - \tau^2 F(\tau/2) = -\tau^2(F(\tau/2) - 4F(2\tau)) = -\tau^2 E_2^*(\tau). \end{aligned}$$

(iii) In $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum \sigma_1(k) e^{2\pi i k\tau}$ the sum tends to 0 as $\operatorname{Im}(\tau) \to \infty$, so by (10) $E_2^*(\tau) \to \frac{\pi^2}{3} - 4\frac{\pi^2}{3} = -\pi^2$.

(iv) We first show

$$E_2^*(1 - 1/\tau) = \tau^2\Big[F\Big(\frac{\tau - 1}{2}\Big) - F(\tau/2)\Big]. \tag{11}$$

F has period 1, and Lemma 3.9(c) can be written $F(w) = w^{-2}[F(-1/w) + 2\pi i w]$. Let $u = (\tau - 1)/2$ and $w = u/\tau = \frac{1}{2} - \frac{1}{2\tau}$. Then $-1/w = -2 - \frac{2}{\tau - 1}$, so $F(-1/w) = F(-1/u) = u^2 F(u) - 2\pi i u$, and

$$F\Big(\frac{1}{2} - \frac{1}{2\tau}\Big) = \frac{\tau^2}{u^2}[u^2 F(u) - 2\pi i u + 2\pi i u/\tau] = \tau^2 F(u) - \frac{2\pi i \tau^2(1 - 1/\tau)}{u} = \tau^2 F\Big(\frac{\tau - 1}{2}\Big) - 4\pi i\tau.$$

Also $4F(2 - 2/\tau) = 4F(-1/(\tau/2)) = 4[(\tau/2)^2 F(\tau/2) - 2\pi i\tau/2] = \tau^2 F(\tau/2) - 4\pi i\tau$. Subtracting gives (11), since $E_2^*(1 - 1/\tau) = F(\frac{1}{2} - \frac{1}{2\tau}) - 4F(2 - \frac{2}{\tau})$. By the expansion of F, $F(\frac{\tau - 1}{2}) - F(\tau/2) = -8\pi^2 \sum_k \sigma_1(k) e^{\pi i k\tau}((-1)^k - 1) = 16\pi^2 e^{\pi i\tau} + O(|e^{3\pi i\tau}|)$, which with (11) gives (iv). For θ^4: (i)–(iii) follow from Section 1 ($\theta(-1/\tau)^4 = (\tau/i)^2\theta(\tau)^4 = -\tau^2\theta(\tau)^4$, $\theta \to 1$), and (iv) from Corollary 1.8: $\theta(1 - 1/\tau)^4 \sim 16(\tau/i)^2 e^{\pi i\tau}$. ∎

Proof of the four-squares theorem. Consider $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, holomorphic in $\mathbb{H}$, invariant under T_2 and S by Proposition 3.8. Since $\theta(\tau)^4 \to 1$ and $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i\tau}(-1)$ as $\operatorname{Im}(\tau) \to \infty$, f is bounded in $\mathcal{F}$, hence in $\mathbb{H}$. By Theorem 3.4, f is constant, equal to $-\pi^2$ by (iii). So $\theta^4 = -\pi^{-2} E_2^*$, which proves the theorem; in particular $r_4(n) \geq 8 > 0$ for all $n \geq 1$. ∎

Appendix A. Asymptotics

Overview

When a function cannot be computed explicitly, one studies its asymptotic behavior; complex analysis is of crucial help. Typically we study, for large s,

$$I(s) = \int_a^b e^{-s\Phi(x)}\,dx. \tag{1}$$

Three guiding principles:

(i) *Deformation of contour.* If Φ is complex-valued, the integrand oscillates rapidly for large s and cancellations mask the behavior. When Φ is holomorphic one may change the contour so that Φ is essentially real on it, and then read off the behavior (illustrated with Bessel functions).

(ii) *Laplace's method.* If Φ is real and $s > 0$, the main contribution comes from near a minimum of Φ, giving an expansion in terms of the quadratic behavior of Φ there (Stirling's formula, the Airy function).

(iii) *Generating functions.* The asymptotics of a sequence F_n can be studied via

$$F_n = \int_\gamma F(e^{2\pi i z})e^{-2\pi i n z}\,dz,$$

γ a segment of unit length in the upper half-plane, a variant of (1); we obtain the asymptotics of the partition function $p(n)$.

1 Bessel functions

Bessel functions arise in problems with rotational symmetry (e.g. the Fourier transform of a radial function on $\mathbb{R}^d$ involves $J_{d/2-1}$). For $\nu > -1/2$,

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}\int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{2}$$

$J_{-1/2}(s) = \lim_{\nu\to -1/2} J_\nu(s) = \sqrt{2/\pi s}\cos s$ and $J_{1/2}(s) = \sqrt{2/\pi s}\sin s$; J_ν is elementary only for half-integral ν. These examples suggest:

Theorem 1.1

$$J_\nu(s) = \sqrt{\frac{2}{\pi s}}\cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2}) \text{ as } s\to\infty.$$

It suffices to study

$$I(s) = \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{3}$$

Let $f(z) = e^{isz}(1-z^2)^{\nu-1/2}$ in the plane slit along $(-\infty,-1]\cup[1,\infty)$, with the branch positive on $(-1,1)$. For $s>0$ fixed, Cauchy's theorem on the rectangle $\gamma_{\epsilon,R}$ with vertices $-1+\epsilon$, $1-\epsilon$, $1-\epsilon+iR$, $-1+\epsilon+iR$ (letting $\epsilon\to 0$, $R\to\infty$) gives

$$I(s) = -I_-(s) - I_+(s),$$

$I_\pm$ the integrals over the vertical half-lines from ± 1 upward, oriented upward for I_+ and downward for I_- appropriately. On the contour of I_+, $z = 1 + iy$:

$$I_+(s) = ie^{is} \int_0^\infty e^{-sy}(1 - (1+iy)^2)^{\nu-1/2}\, dy, \tag{4}$$

and similarly for I_-. In (3), e^{isx} oscillates rapidly; in (4), e^{-sy} decreases rapidly except near $y = 0$, so the main contribution comes from y near 0:

Proposition 1.2

Suppose a and m are fixed, with $a > 0$ and $m > -1$. Then as $s \to \infty$

$$\int_0^a e^{-sx} x^m\, dx = s^{-m-1}\Gamma(m+1) + O(e^{-cs}), \tag{5}$$

for some positive c.

Proof. $m > -1$ guarantees convergence at 0. Write $\int_0^a = \int_0^\infty - \int_a^\infty$. The first equals $s^{-m-1}\Gamma(m+1)$ (change variables $x \mapsto x/s$). For the second,

$$\int_a^\infty e^{-sx} x^m\, dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m\, dx = O(e^{-cs}) \tag{6}$$

whenever $c < a$. ∎

Proof of Theorem 1.1. $(1 - (1+iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2})$ for $0 \le y \le 1$, and $= O(y^{\nu-1/2} + y^{2\nu-1})$ for $y \ge 1$. Proposition 1.2 with $a = 1$, $m = \nu - 1/2$ (and $\nu + 1/2$), together with (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2}),$$
$$I_-(s) = i(2i)^{\nu-1/2} e^{-is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2})$$

(with the appropriate sign conventions). Since $J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)]$ and $\Gamma(1/2) = \sqrt{\pi}$, combining the two conjugate terms gives the theorem: $(-2i)^{\nu-1/2}$ and $(2i)^{\nu-1/2}$ contribute the phases $e^{\mp i\pi(\nu-1/2)/2}$, and the sum is $2^{\nu+1/2} s^{-\nu-1/2}\Gamma(\nu+\frac{1}{2})\cos(s - \frac{\pi\nu}{2} - \frac{\pi}{4})$ up to $O(s^{-\nu-3/2})$. ∎

Under restricted circumstances Proposition 1.2 extends to $\mathrm{Re}(s) \ge 0$:

Proposition 1.3

Suppose a and m are fixed, with $a > 0$ and $-1 < m < 0$. Then as $|s| \to \infty$ with $\mathrm{Re}(s) \ge 0$,

$$\int_0^a e^{-sx} x^m\, dx = s^{-m-1}\Gamma(m+1) + O(1/|s|).$$

(Here s^{-m-1} is the branch that is positive for $s > 0$.)

Proof. First, for $\mathrm{Re}(s) \ge 0$, $s \ne 0$, $\int_0^\infty e^{-sx} x^m\, dx = \lim_{N\to\infty} \int_0^N e^{-sx} x^m\, dx$ exists and equals $s^{-m-1}\Gamma(m+1)$. Write $\int_0^N = \int_0^a + \int_a^N$; the first is entire in s. Since $-\frac{1}{s}\frac{d}{dx}(e^{-sx}) = e^{-sx}$, integration by parts gives

$$\int_a^N e^{-sx} x^m\, dx = -\frac{m}{s}\int_a^N e^{-sx} x^{m-1}\, dx - \Big[\frac{e^{-sx}x^m}{s}\Big]_a^N. \tag{7}$$

As $m < 0$, $\int_a^\infty x^{m-1} dx < \infty$, so letting $N \to \infty$, $\int_a^\infty e^{-sx} x^m\, dx$ is analytic in $\mathrm{Re}(s) > 0$ and continuous on $\mathrm{Re}(s) \ge 0$, $s \ne 0$; it equals $s^{-m-1}\Gamma(m+1) - \int_0^a$ for $s > 0$, hence everywhere there. Finally $\int_0^a = \int_0^\infty - \int_a^\infty$, and by (7), $\int_a^\infty e^{-sx} x^m\, dx = O(1/|s|)$. ∎

(Better error terms, or other ranges of m, require smooth cut-offs to mitigate the end-point $x = a$.)

2 Laplace's method; Stirling's formula

Motivation. Integrals $\int e^{-s\Phi(x)}\psi(x)dx$ with large s are dominated by a neighbourhood of the minimum of Φ. Laplace's method makes this precise, and applied to $\Gamma(s)$ it gives Stirling's formula.

Consider $\int_a^b e^{-s\Phi(x)}\psi(x)\,dx$ with real Φ, and Φ, ψ indefinitely differentiable. Assume there is $x_0 \in (a,b)$ with $\Phi'(x_0) = 0$ and $\Phi''(x) > 0$ throughout $[a,b]$.

Proposition 2.1 (Laplace's method)

Under the above assumptions, with $s > 0$ and $s \to \infty$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = e^{-s\Phi(x_0)}\Big[\frac{A}{s^{1/2}} + O\Big(\frac{1}{s}\Big)\Big], \qquad A = \frac{(2\pi)^{1/2}\psi(x_0)}{(\Phi''(x_0))^{1/2}}. \tag{8}$$

Proof. Replacing Φ by $\Phi - \Phi(x_0)$, assume $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, $\frac{\Phi(x)}{(x-x_0)^2} = \frac{\Phi''(x_0)}{2}\varphi(x)$ with φ smooth, $\varphi(x) = 1 + O(x - x_0)$. Near x_0 change variables $y = (x - x_0)(\varphi(x))^{1/2}$; then $dx/dy = 1 + O(y)$ and $\psi(x) = \tilde{\psi}(y) = \psi(x_0) + O(y)$. On a small interval $[a', b'] \ni x_0$,

$$\int_{a'}^{b'} e^{-s\Phi(x)}\psi(x)\,dx = \psi(x_0)\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}\,dy + O\Big(\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}|y|\,dy\Big), \tag{9}$$

$\alpha < 0 < \beta$. With $y^2 = X$, $dy = \frac{1}{2}X^{-1/2}dX$, and (5), the first integral is $s^{-1/2}(\frac{2\pi}{\Phi''(x_0)})^{1/2} + O(e^{-\delta s})$; similarly the second is $O(1/s)$. On $[a, a']$ and $[b', b]$, $\Phi \geq c > 0$, so those integrals decay exponentially. ∎

(8) extends to complex s with $\mathrm{Re}(s) \geq 0$, using integration by parts to handle the oscillations of $e^{-s\Phi}$:

Proposition 2.2

With the same assumptions on Φ and ψ, the relation (8) continues to hold if $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$.

Proof. Proceed to (9); the first term is handled by Proposition 1.3 with $m = -1/2$. Observation: if $\Phi \geq 0$ and $|\Phi'(x)| \geq c > 0$ on $[a,b]$, then for $\mathrm{Re}(s) \geq 0$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = O\Big(\frac{1}{|s|}\Big) \qquad \text{as } |s| \to \infty; \tag{10}$$

indeed the integral is $-\frac{1}{s}\int_a^b \frac{d}{dx}(e^{-s\Phi})\frac{\psi}{\Phi'}\,dx = \frac{1}{s}\int_a^b e^{-s\Phi}\frac{d}{dx}\Big(\frac{\psi}{\Phi'}\Big)dx - \frac{1}{s}\Big[e^{-s\Phi}\frac{\psi}{\Phi'}\Big]_a^b$, and $|e^{-s\Phi}| \leq 1$. This handles $[a, a']$ and $[b', b]$, where $|\Phi'| \geq c > 0$ (as $\Phi'(x_0) = 0$ and $\Phi'' \geq c_1 > 0$). The second term of (9) has the form $\int e^{-s\frac{\Phi''(x_0)}{2}y^2}y\eta(y)\,dy$ with η differentiable; write it as $\frac{-1}{s\Phi''(x_0)}\int \frac{d}{dy}(e^{-s\frac{\Phi''(x_0)}{2}y^2})\eta(y)\,dy$ and integrate by parts to get $O(1/|s|)$. ∎

For purely imaginary $s = it$ this is the **method of stationary phase**; points with $\Phi'(x_0) = 0$ are **critical points**.

For $\delta > 0$ let $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and let $\log s$ be the principal branch.

Theorem 2.3 (Stirling's formula)

If $|s| \to \infty$ with $s \in S_\delta$, then

$$\Gamma(s) = e^{s\log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \Big(1 + O\Big(\frac{1}{|s|^{1/2}}\Big)\Big). \tag{11}$$

(With more effort the error term improves to $O(1/|s|)$, with a full asymptotic expansion in powers of $1/s$. Note $\Gamma(s) \sim \sqrt{2\pi} s^{s-1/2} e^{-s}$, the usual form.)

Proof. First let $\mathrm{Re}(s) \geq 0$, with uniform error terms off a neighborhood of 0. For $s > 0$,

$$\Gamma(s) = \int_0^\infty e^{-x} x^s \frac{dx}{x} = \int_0^\infty e^{-x + s\log x} \frac{dx}{x} = e^{s\log s} e^{-s} \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}, \qquad \Phi(x) = x - 1 - \log x,$$

after $x \mapsto sx$; by analytic continuation $\Gamma(s) = e^{s\log s} e^{-s} I(s)$ for $\mathrm{Re}(s) > 0$, $I(s) = \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}$. It suffices to show

$$I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\Big(\frac{1}{|s|}\Big) \qquad \text{for } \mathrm{Re}(s) \geq 0. \tag{12}$$

$\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$, $\Phi''(1) = 1$; so Φ is convex, positive, with minimum at 1. Apply Proposition 2.2 on $[1/2, 2]$ with $x_0 = 1$, $\psi(x) = 1/x$: this gives (12) there. On $[0, 1/2]$, since $\Phi'(x) = 1 - 1/x$,

$$\int_\epsilon^{1/2} e^{-s\Phi(x)} \frac{dx}{x} = -\frac{1}{s} \int_\epsilon^{1/2} \frac{d}{dx}(e^{-s\Phi(x)}) \frac{dx}{\Phi'(x)x} = -\frac{1}{s}\Big[\frac{e^{-s\Phi(x)}}{x-1}\Big]_\epsilon^{1/2} - \frac{1}{s}\int_\epsilon^{1/2} e^{-s\Phi(x)} \frac{dx}{(x-1)^2}.$$

Since $\Phi(\epsilon) \to +\infty$ as $\epsilon \to 0$ and $|e^{-s\Phi}| \leq 1$, in the limit

$$\int_0^{1/2} e^{-s\Phi} \frac{dx}{x} = \frac{2}{s} e^{-s\Phi(1/2)} - \frac{1}{s}\int_0^{1/2} e^{-s\Phi} \frac{dx}{(x-1)^2} = O(1/|s|).$$

$\int_2^\infty$ is treated analogously, since $\int_2^\infty (x-1)^{-2} dx < \infty$. So (11) holds for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

To pass to $\mathrm{Re}(s) \leq 0$, $s \in S_\delta$: for $\mathrm{Re}(s) \geq 0$, $s = \sigma + it$, $t \neq 0$, $\log(-s) = \log s - i\pi$ if $t > 0$ and $\log s + i\pi$ if $t < 0$. Hence for $G(s) = e^{s\log s} e^{-s}$,

$$G(-s)^{-1} = e^{s\log s} e^{-s} e^{-si\pi} \ (t > 0), \qquad G(-s)^{-1} = e^{s\log s} e^{-s} e^{si\pi} \ (t < 0). \tag{13}$$

Also

$$\Gamma(s)\Gamma(-s) = \frac{-\pi}{s \sin \pi s}, \tag{14}$$

from $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ and $\Gamma(1-s) = -s\Gamma(-s)$ (Theorem 1.4 of Chapter 6, Lemma 1.2 of Chapter 6). Combining (13), (14) and $(1 + O(|s|^{-1/2}))^{-1} = 1 + O(|s|^{-1/2})$ extends (11) to S_δ (in S_δ, $\sin \pi s$ is comparable to $e^{\pm i\pi s}/(\mp 2i)$ for large $|t|$, and near the negative real axis s is excluded). ■

3 The Airy function

The Airy function appeared in optics (intensity of light near a caustic):

$$\mathrm{Ai}(s) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{i(x^3/3 + sx)} \, dx, \qquad s \in \mathbb{R}. \tag{15}$$

The integral converges and is continuous in s: $\frac{1}{i(x^2+s)} \frac{d}{dx} e^{i(x^3/3+sx)} = e^{i(x^3/3+sx)}$, so for $a \geq 2|s|^{1/2}$,

$$\int_a^R e^{i(x^3/3+sx)} \, dx = \int_a^R \frac{1}{i(x^2+s)} \frac{d}{dx} e^{i(x^3/3+sx)} \, dx, \tag{16}$$

and integrating by parts and letting $R \to \infty$ shows uniform convergence for $|s| \le a^2/4$; similarly on $(-\infty, -a]$. Replacing $\mathbb{R}$ by $L_\delta = \{x + i\delta : x \in \mathbb{R}\}$, $\delta > 0$, improves matters: by Cauchy's theorem on rectangles γ_R with $f(z) = e^{i(z^3/3+sz)}$, which is $O(e^{-\delta x^2})$ on L_δ and $O(e^{-yR^2})$ on the vertical sides,

$$\mathrm{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} e^{i(z^3/3+sz)}\, dz,$$

and since $f(z) = O(e^{-\delta x^2})$ on L_δ for every complex s, Ai extends to an entire function. It satisfies

$$\mathrm{Ai}''(s) = s\,\mathrm{Ai}(s) : \tag{17}$$

$\mathrm{Ai}''(s) - s\,\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{L_\delta}(-z^2 - s)e^{i(z^3/3+sz)}dz = \frac{i}{2\pi}\int_{L_\delta}\frac{d}{dz}(f(z))\,dz = 0$, since f vanishes at infinity along L_δ. Comparing (17) with $y'' = Ay$ (A large) suggests $\mathrm{Ai}(s)$ decreases rapidly as $s \to \infty$ (it stays bounded by (16), excluding $e^{\sqrt{A}s}$) and oscillates as $s \to -\infty$.

Theorem 3.1

Suppose $u > 0$. Then as $u \to \infty$, (i) $\mathrm{Ai}(-u) = \pi^{-1/2}u^{-1/4}\cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4}))$.
(ii) $\mathrm{Ai}(u) = \dfrac{1}{2\pi^{1/2}}u^{-1/4}e^{-\frac{2}{3}u^{3/2}}(1 + O(1/u^{3/4}))$.

Proof. (i) Change variables $x \mapsto u^{1/2}x$ with $s = -u$: $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$, where

$$I_-(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{it(x^3/3-x)}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx \tag{19}$$

with $s = -it$, $\Phi(x) = x^3/3 - x$. $\Phi'(x) = x^2 - 1$: two critical points $x_0 = \pm 1$; $\Phi''(x) = 2x$, $\Phi(\pm 1) = \mp 2/3$. Split into $[-2, 0]$, $[0, 2]$ (one critical point each) and $(-\infty, -2]$, $[2, \infty)$. On $[0, 2]$, Proposition 2.2 with $s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi(1) = -2/3$, $\Phi''(1) = 2$ gives $\frac{1}{2\sqrt{\pi}}e^{-i\frac{2}{3}t}\left[\frac{1}{(-it)^{1/2}} + O(\frac{1}{|t|})\right]$; the interval $[-2, 0]$ gives $\frac{1}{2\sqrt{\pi}}e^{i\frac{2}{3}t}\left[\frac{1}{(it)^{1/2}} + O(\frac{1}{|t|})\right]$ (there Φ has a maximum; apply the proposition to $-\Phi$ with $s = it$). The complementary integrals are $O(1/|t|)$ by integration by parts ($\Phi' \neq 0$ there), e.g. $\int_{-\infty}^{-2} e^{it\Phi}dx = \lim_N \frac{1}{it}\int_{-N}^{-2}\frac{d}{dx}(e^{it\Phi})\frac{dx}{\Phi'(x)}$. Adding and inserting into $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$ gives (i).

(ii) With $s = u$, $\mathrm{Ai}(u) = u^{1/2}I_+(u^{3/2})$,

$$I_+(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx, \qquad F(x) = -i(x^3/3 + x). \tag{20}$$

There is no real critical point; repeated integration by parts shows rapid decay, but not its exact order. Follow the Bessel-function principle: deform the line to a contour γ on which (a) $\mathrm{Im}(F) = 0$, and (b) $\mathrm{Re}(F)$ has a non-degenerate minimum at some z_0 (second derivative along γ positive); then $F'(z_0) = 0$. Through such z_0 pass two orthogonal curves γ_1, γ_2 on which F is real, $\mathrm{Re}(F)|_{\gamma_1}$ having a minimum and $\mathrm{Re}(F)|_{\gamma_2}$ a maximum at z_0; take $\gamma = \gamma_1$—the **method of steepest descent** ($-\mathrm{Re}\,F$ has a saddle at z_0). Here $\mathrm{Re}(F) = x^2y - y^3/3 + y$, $\mathrm{Im}(F) = -x^3/3 + xy^2 - x$, $F'(z) = -i(z^2 + 1)$, critical points $z_0 = \pm i$. At $z_0 = i$ the curves with $\mathrm{Im}(F) = 0$ are $\gamma_1 = \{y^2 = x^2/3 + 1\}$ and $\gamma_2 = \{x = 0\}$; on γ_2, $\mathrm{Re}(F)$ has a maximum at i, so take $\gamma = \gamma_1$, the branch $y = (x^2/3 + 1)^{1/2}$ of a hyperbola, asymptotic to the rays $re^{i\pi/6}$, $re^{i5\pi/6}$. Then

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx = \frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz, \tag{21}$$

by Cauchy's theorem on the region between $\mathbb{R}$ and γ cut off by the circle of radius R (there $e^{-sF(z)} = O(e^{-cyx^2})$, so the arcs contribute $O(\int_0^{\pi/6} e^{-cR^2\sin\theta}R\,d\theta) = O(1/R)$). On γ, $\Phi(x) = \mathrm{Re}(F) = y(x^2 - y^2/3 + 1) = (\frac{8}{9}x^2 + \frac{2}{3})(x^2/3 + 1)^{1/2}$, and $dz = dx + i\frac{x}{3}(x^2/3 + 1)^{-1/2}dx$; since Φ is even and $x(x^2/3 + 1)^{-1/2}$ odd,

$$\frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx. \tag{22}$$

As $(1+u)^{1/2} = 1 + u/2 + O(u^2)$, $\Phi(x) = \frac{2}{3} + x^2 + O(x^4)$, so $\Phi(0) = 2/3$, $\Phi''(0) = 2$. Proposition 2.1 on $[-c, c]$ with $\psi = 1/2\pi$ gives $e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\big[\frac{1}{s^{1/2}} + O(\frac{1}{s})\big]$; the rest is $O(e^{-2s/3}e^{-\delta s})$, since there $\Phi(x) \geq \frac{2}{3} + c_1 x^2$. So $I_+(s) = e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\big[\frac{1}{s^{1/2}} + O(\frac{1}{s})\big]$, which gives (ii). ∎

4 The partition function

Motivation. How many ways can n be written as a sum of positive integers? The generating function of $p(n)$ is a product of geometric series, and the saddle-point methods of this appendix give its asymptotic size.

Theorem 4.1 (Hardy–Ramanujan)

If p denotes the partition function, then (i) $p(n) \sim \dfrac{1}{4\sqrt{3}\,n} e^{Kn^{1/2}}$ as $n \to \infty$, where $K = \pi\sqrt{2/3}$.
(ii) A much more precise assertion is that

$$p(n) = \frac{1}{2\pi\sqrt{2}}\frac{d}{dn}\Big(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

(Since $(n-\frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, $e^{K(n-1/24)^{1/2}} \sim e^{Kn^{1/2}}$, and (ii) implies (i).)

Heuristic principle: if $\sum F_n w^n$ has the unit disc as disc of convergence, the asymptotic behavior of F_n is determined by the location and nature of the "singularities" of F on the unit circle, each contributing according to its "order". (E.g. if F is meromorphic in a larger disc with only a pole of order r at $w = 1$ on the circle, then $F_n = P(n) + O(e^{-\epsilon n})$ for a polynomial P of degree $r-1$, $\sum P(n)w^n$ being essentially the principal part.)

By Euler's formula (Theorem 2.1), with $w = e^{2\pi i z}$, $z = x + iy$, $y > 0$,

$$\sum_{n=0}^{\infty} p(n)e^{2\pi i n z} = f(z) = \prod_{n=1}^{\infty}\frac{1}{1-e^{2\pi i n z}}, \qquad p(n) = \int_\gamma f(z)e^{-2\pi i n z}\,dz, \tag{23}$$

γ the segment from $-1/2 + i\delta$ to $1/2 + i\delta$, $\delta > 0$ to be chosen in terms of n. f is largest near $z = 0$: $|f(x+iy)| \leq f(iy)$ (positive coefficients), increasing as y decreases. Fortunately f is a variant of the Dedekind eta function $\eta(z) = e^{\pi i z/12}\prod(1-e^{2\pi i n z})$: $f(z) = e^{\pi i z/12}\eta(z)^{-1}$ (the $1/12$ explains the $1/24$ in the formula). By $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$ (Proposition 1.9),

$$f(z) = \sqrt{z/i}\,e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}f(-1/z), \tag{24}$$

and as $\mathrm{Im}(-1/z) \to \infty$, $f(-1/z) \to 1$ rapidly, since

$$f(z) = 1 + O(e^{-2\pi y}), \qquad z = x + iy,\ y \geq 1. \tag{25}$$

So $f_1(z) = \sqrt{z/i}\,e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}$ approximates f near 0, and by (24) $p(n) = p_1(n) + E(n)$ with

$$p_1(n) = \int_\gamma \sqrt{z/i}\,e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}e^{-2\pi i n z}\,dz, \qquad E(n) = \int_\gamma \sqrt{z/i}\,e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}e^{-2\pi i n z}(f(-1/z)-1)\,dz.$$

The error term. For $z \in \gamma$, since $\mathrm{Re}(i/z) = \delta/(\delta^2 + x^2)$,

$$\Big|\sqrt{z/i}\,e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}e^{-2\pi i n z}\Big| \leq c e^{2\pi n\delta}e^{\frac{\pi}{12}\frac{\delta}{\delta^2+x^2}}. \tag{26}$$

By (25) with z replaced by $-1/z$,

$$|f(-1/z)-1| \le c e^{-2\pi \frac{\delta}{\delta^2+x^2}} \qquad \text{if } \frac{\delta}{\delta^2+x^2} \ge 1; \tag{27}$$

and since $|f(z)| \le f(iy) \le C e^{\frac{\pi}{12y}}$ for $y \le 1$ (by (24)),

$$|f(-1/z)-1| = O\big(e^{\frac{\pi}{12}\frac{\delta^2+x^2}{\delta}}\big) = O\big(e^{\frac{\pi}{48\delta}}\big) \qquad \text{if } \frac{\delta}{\delta^2+x^2} \le 1, \tag{28}$$

as $|x| \le 1/2$. Using (26), (27) when $\frac{\delta}{\delta^2+x^2} \ge 1$ gives $O(e^{2\pi n\delta})$ (as $2\pi > \pi/12$); using (26), (28) otherwise gives $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Choose δ to minimize, $2\pi n\delta = \frac{\pi}{48\delta}$, i.e. $\delta = \frac{1}{4\sqrt{6}} n^{-1/2}$; then

$$E(n) = O\big(e^{\frac{\pi}{\sqrt{6}} n^{1/2}}\big) = O\big(e^{\frac{K}{2} n^{1/2}}\big).$$

The main term. Add to γ the end-segments $[-1/2, -1/2+i\delta]$ and $[1/2+i\delta, 1/2]$, forming Γ; on them $\sqrt{z/i}\, e^{\frac{\pi i}{12z}} = O(1)$, so they contribute $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4} n^{1/2}})$, absorbed in $E(n)$. So

$$p_1(n) = \int_\Gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz. \tag{29}$$

Change variables $z \mapsto \mu z$ so that the exponentials combine to $e^{Ai(\frac{1}{z}-z)}$: this requires $A = 2\pi\mu(n-\frac{1}{24})$ and $A = \frac{\pi}{12\mu}$, i.e.

$$A = \frac{\pi}{\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{1/2}, \qquad \mu = \frac{1}{2\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{-1/2}.$$

Then

$$p_1(n) = \mu^{3/2} \int_{\Gamma'} e^{-sF(z)} \sqrt{z/i}\, dz, \qquad F(z) = i(z-1/z), \quad s = A, \tag{30}$$

$\Gamma' = \mu^{-1}\Gamma$ the union of $[-a_n, -a_n+i\delta']$, $[-a_n+i\delta', a_n+i\delta']$, $[a_n+i\delta', a_n]$, with $a_n = \frac{1}{2}\mu^{-1} \approx \sqrt{6}\, n^{1/2}$, $\delta' = \mu^{-1}\delta \to 1$. Steepest descent: F has the critical point $z = i$ in $\mathbb{H}$; through it, F is real on the imaginary axis (maximum, rejected) and on the unit circle (minimum at i). By Cauchy's theorem replace Γ' by Γ'': the segments $[-a_n, -1]$, $[1, a_n]$ and the upper unit semicircle from -1 to 1. On the segments F is purely imaginary, so they contribute $O(a_n^{3/2}\mu^{3/2}) = O(1)$. On the semicircle, $z = e^{i\theta}$ (θ from π to 0), $dz = ie^{i\theta}d\theta$, $i(z-1/z) = -2\sin\theta$ and $\sqrt{z/i} = e^{i(\theta/2-\pi/4)}$; with $\theta = \varphi + \pi/2$ the integrand becomes $-e^{2s\cos\varphi} e^{3i\varphi/2}$, and the contribution is

$$\mu^{3/2} \int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\big(\cos(3\varphi/2) + i\sin(3\varphi/2)\big)\, d\varphi.$$

Laplace's method with $\Phi(\varphi) = -\cos\varphi$, $\varphi_0 = 0$, $\Phi(0) = -1$, $\Phi''(0) = 1$, $\psi(\varphi) = \cos(3\varphi/2) + i\sin(3\varphi/2)$, $\psi(0) = 1$, gives $\mu^{3/2} e^{2s} \frac{\sqrt{2\pi}}{(2s)^{1/2}}(1 + O(s^{-1/2}))$. Since $2s = K(n-\frac{1}{24})^{1/2}$ and $\mu = \frac{1}{2\sqrt{6}}(n-\frac{1}{24})^{-1/2}$,

$$p(n) = \frac{1}{4\sqrt{3}\, n} e^{Kn^{1/2}}(1 + O(n^{-1/4})),$$

which is (i).

The precise formula (ii). Write $p_1(n) = \frac{d}{dn} q(n) + e(n)$, where

$$q(n) = \frac{1}{2\pi} \int_\Gamma (z/i)^{-1/2} e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz$$

(differentiating $e^{-2\pi i n z}$ in n produces $-2\pi i z$, and $\frac{-2\pi i z}{2\pi}(z/i)^{-1/2} = (z/i)^{1/2}$), and $e(n)$ comes from the dependence of $\Gamma = \Gamma_n$ on n; by Cauchy's theorem it is $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4} n^{1/2}})$. As before,

$$q(n) = \frac{\mu^{1/2}}{2\pi} \int_{\Gamma''} e^{-sF(z)} (z/i)^{-1/2}\, dz, \tag{31}$$

the segments contributing $O(a_n^{1/2}\mu^{1/2}) = O(1)$ to $\frac{d}{dn}q(n)$. On the semicircle the integral equals

$$\frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\big(\cos(\varphi/2) + i\sin(\varphi/2)\big)\,d\varphi = \frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\cos(\varphi/2)\,d\varphi,$$

the odd part vanishing. With $\cos\varphi = 1 - 2\sin^2(\varphi/2)$ and $x = \sin(\varphi/2)$, $\cos(\varphi/2)d\varphi = 2dx$, this is

$$\frac{\mu^{1/2}}{\pi}e^{2s}\int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\,dx, \qquad \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\,dx = \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}),$$

and the same holds after differentiating in s. Collecting the error terms,

$$p(n) = \frac{d}{dn}\Big(\frac{\mu^{1/2}e^{2s}}{2\sqrt{\pi}\,s^{1/2}}\Big) + O\big(e^{\frac{K}{2}n^{1/2}}\big).$$

Since $\mu^{1/2}/s^{1/2} = (2\pi)^{-1/2}(n-\frac{1}{24})^{-1/2}$ and $2s = K(n-\frac{1}{24})^{1/2}$, this is (ii).

5 Selected exercises

Motivation. Two problems from Stein–Shakarchi (numbers kept): a smooth cut-off upgrades the $O(1/|s|)$ error of Proposition 1.3 to $O(|s|^{-N})$ for all $m > -1$, and the residue theorem makes the heuristic principle of Section 4 exact for meromorphic generating functions.

Exercise A.1 (Smooth cut-offs)

Let η be an indefinitely differentiable function on $\mathbb{R}$, supported in a finite interval, with $\eta(x) = 1$ for x near 0. Show that if $m > -1$ and $N > 0$, then

$$\int_0^\infty e^{-sx}x^m\eta(x)\,dx = s^{-m-1}\Gamma(m+1) + O(|s|^{-N})$$

as $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$ (principal branch of s^{-m-1}).

Solution. Fix $\delta, b > 0$ with $\eta = 1$ on $[0,\delta]$ and $\eta = 0$ on $[b,\infty)$; we may take N to be an integer. Throughout $\mathrm{Re}(s) \geq 0$, $s \neq 0$, so $|e^{-sx}| \leq 1$ for $x \geq 0$, and $e^{-sx} = -\frac{1}{s}\frac{d}{dx}e^{-sx}$.

Step 1. (Functions vanishing near 0.) If φ is indefinitely differentiable with compact support in $(0,\infty)$, then N integrations by parts (no boundary terms) give

$$\Big|\int_0^\infty e^{-sx}\varphi(x)\,dx\Big| = \Big|s^{-N}\int_0^\infty e^{-sx}\varphi^{(N)}(x)\,dx\Big| \leq \|\varphi^{(N)}\|_{L^1}|s|^{-N}.$$

Step 2. (Case $m = 0$.) One integration by parts gives

$$\int_0^\infty e^{-sx}\eta(x)\,dx = \Big[-\frac{e^{-sx}\eta(x)}{s}\Big]_0^\infty + \frac{1}{s}\int_0^\infty e^{-sx}\eta'(x)\,dx = \frac{1}{s} + O(|s|^{-N-1})$$

by Step 1 applied to $\varphi = \eta'$, which vanishes on $[0,\delta]$ and on $[b,\infty)$. Since $\Gamma(1) = 1$ this is the claim.

Step 3. (Case $-1 < m < 0$.) By the proof of Proposition 1.3,

$$\int_0^\infty e^{-sx}x^m\,dx = \lim_{R\to\infty}\int_0^R e^{-sx}x^m\,dx = s^{-m-1}\Gamma(m+1).$$

So, with $g(x) = x^m(1-\eta(x))$, it suffices to show

$$J(s) = \lim_{R\to\infty}\int_0^R e^{-sx}g(x)\,dx = O(|s|^{-N}).$$

Now g is indefinitely differentiable on $[0,\infty)$, vanishes on $[0,\delta]$, and equals x^m on $[b,\infty)$. Hence $g^{(j)}(R) = O(R^{m-j}) \to 0$ for all $j \ge 0$, and $g^{(j)} \in L^1(0,\infty)$ for $j \ge 1$ since $m - j < -1$. One integration by parts gives

$$\int_0^R e^{-sx} g\,dx = -\frac{e^{-sR}g(R)}{s} + \frac{1}{s}\int_0^R e^{-sx} g'\,dx \;\longrightarrow\; \frac{1}{s}\int_0^\infty e^{-sx} g'(x)\,dx \quad (R\to\infty).$$

Repeating with $g', \dots, g^{(N-1)}$ (the boundary terms at R tend to 0, those at 0 vanish), $J(s) = s^{-N}\int_0^\infty e^{-sx} g^{(N)}(x)\,dx$, so $|J(s)| \le \|g^{(N)}\|_{L^1}|s|^{-N}$.

Step 4. (General $m > -1$.) Choose the integer $k \ge 0$ with $k-1 < m \le k$ and put $\mu = m - k \in (-1, 0]$. For $0 \le j \le k-1$, $(x^m)^{(j)}$ is a multiple of x^{m-j} with $m - j > 0$, so it vanishes at 0; hence k integrations by parts (support of η bounded) give

$$\int_0^\infty (x^m)^{(k)} e^{-sx}\eta\,dx = (-1)^k \int_0^\infty x^m \left(e^{-sx}\eta\right)^{(k)} dx.$$

By Leibniz, $(e^{-sx}\eta)^{(k)} = (-s)^k e^{-sx}\eta + \sum_{j=1}^{k}\binom{k}{j}(-s)^{k-j}e^{-sx}\eta^{(j)}$. For $j \ge 1$, $x^m\eta^{(j)}$ has compact support in $(0,\infty)$, so by Step 1 (with $N+k$ in place of N) its contribution is $O(|s|^{k-j}|s|^{-N-k}) = O(|s|^{-N})$. With $c = m(m-1)\cdots(m-k+1)$ we have $(x^m)^{(k)} = c\,x^{\mu}$, and Steps 2–3 give

$$s^k \int_0^\infty e^{-sx} x^m \eta\,dx = c\,s^{-\mu-1}\Gamma(\mu+1) + O(|s|^{-N}).$$

Step 5. (Conclusion.) Divide by s^k: since $s^{-\mu-1}s^{-k} = s^{-m-1}$ for the principal branch and $c\,\Gamma(\mu+1) = \Gamma(m+1)$ by k applications of $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^\infty e^{-sx} x^m \eta(x)\,dx = s^{-m-1}\Gamma(m+1) + O(|s|^{-N-k}),$$

which is stronger than claimed. ∎

Exercise A.4 (Poles on the unit circle)

Suppose $F(w) = \sum_{n\ge 0} F_n w^n$ is meromorphic in a region containing the closed unit disc, and its only poles there lie on the unit circle, at $\alpha_1, \dots, \alpha_k$, of orders $r_1, \dots, r_k$. Let

$$P_j(n) = \frac{1}{(r_j-1)!}\left(\frac{d}{dw}\right)^{r_j-1}\left[(w-\alpha_j)^{r_j} w^{-n-1} F(w)\right]_{w=\alpha_j}.$$

(a) Show that for some $\epsilon > 0$, $F_n = -\sum_{j=1}^{k} P_j(n) + O(e^{-\epsilon n})$ as $n \to \infty$.
(b) Show that $P_j(n) = \alpha_j^{-n} Q_j(n)$ with Q_j a polynomial of degree $r_j - 1$.
(c) Find F_n exactly for $F(w) = \frac{1}{(1-w)(1-w^2)}$, and check it against the number of solutions of $a + 2b = n$ in integers $a, b \ge 0$.

Solution.

(a)

Step 1. (A larger circle.) Let Ω be the region. The closed unit disc is a compact subset of Ω, so $\{|w| \le R_0\} \subset \Omega$ for some $R_0 > 1$. The poles of F are discrete, so only finitely many lie in $\{1 < |w| \le R_0\}$; choose $R \in (1, R_0)$ smaller than all their moduli. Then F is holomorphic on $\{|w| \le R\}$ except at $\alpha_1, \dots, \alpha_k$.

Step 2. (Residues.) Put $G(w) = w^{-n-1}F(w)$. At 0, F is holomorphic with Taylor coefficients F_n, so $\operatorname{res}_0 G = F_n$. At $\alpha_j \ne 0$, w^{-n-1} is holomorphic, so G has a pole of order at most r_j; $(w-\alpha_j)^{r_j}G$

is holomorphic near α_j, and its Taylor coefficient of order r_j-1 is the coefficient of $(w-\alpha_j)^{-1}$ in G. Hence $\operatorname{res}_{\alpha_j} G = P_j(n)$. The residue theorem on $|w|=R$ gives

$$F_n + \sum_{j=1}^{k} P_j(n) = \frac{1}{2\pi i}\oint_{|w|=R} w^{-n-1}F(w)\,dw.$$

Step 3. (Estimate.) With $M = \max_{|w|=R}|F|$, the right side is at most $R\cdot R^{-n-1}M = MR^{-n} = Me^{-\epsilon n}$ with $\epsilon = \log R > 0$.

(The sign matters: for $F = 1/(1-w)$ one has $F_n = 1$ and $P_1(n) = -1$.)

(b) Let $h(w) = (w-\alpha_j)^{r_j}F(w)$, holomorphic near α_j with $h(\alpha_j)\neq 0$ because the pole has order exactly r_j. Since $\left(\frac{d}{dw}\right)^i w^{-n-1} = (-1)^i(n+1)\cdots(n+i)\,w^{-n-1-i}$, Leibniz's rule gives

$$P_j(n) = \frac{\alpha_j^{-n}}{(r_j-1)!}\sum_{i=0}^{r_j-1}\binom{r_j-1}{i}(-1)^i(n+1)\cdots(n+i)\,\alpha_j^{-1-i}h^{(r_j-1-i)}(\alpha_j).$$

The sum is a polynomial $Q_j(n)$; only $i=r_j-1$ contributes to degree r_j-1, with leading coefficient $(-1)^{r_j-1}\alpha_j^{-r_j}h(\alpha_j)/(r_j-1)!\neq 0$. So $P_j(n) = \alpha_j^{-n}Q_j(n)$, and since $|\alpha_j|=1$, $|P_j(n)|$ grows like n^{r_j-1}.

(c)

Step 1. $F(w) = \frac{1}{(1-w)^2(1+w)}$ is rational with poles 1 (order 2) and -1 (order 1). Here $w^{-n-1}F(w) = O(R^{-n-4})$ on $|w|=R$, so the integral in (a) tends to 0 as $R\to\infty$, and $F_n = -P_1(n) - P_2(n)$ exactly.

Step 2. At $\alpha=-1$: $P_2(n) = \left[\frac{w^{-n-1}}{(1-w)^2}\right]_{w=-1} = \frac{(-1)^{n+1}}{4}$.

Step 3. At $\alpha=1$: $(w-1)^2w^{-n-1}F(w) = \frac{w^{-n-1}}{1+w}$, so

$$P_1(n) = \Big[\frac{-(n+1)w^{-n-2}}{1+w} - \frac{w^{-n-1}}{(1+w)^2}\Big]_{w=1} = -\frac{n+1}{2} - \frac{1}{4} = -\frac{2n+3}{4}.$$

Step 4. Hence $F_n = \frac{2n+3}{4} + \frac{(-1)^n}{4}$, i.e. $F_n = \frac{n}{2}+1$ for even n and $\frac{n+1}{2}$ for odd n; in both cases $F_n = \lfloor n/2\rfloor + 1$. On the other hand $F(w) = \sum_{a\geq 0} w^a \sum_{b\geq 0} w^{2b}$, so F_n counts the pairs (a,b) with $a+2b=n$, i.e. $b = 0,1,\ldots,\lfloor n/2\rfloor$: again $\lfloor n/2\rfloor+1$. ∎

Appendix B. Simple Connectivity and Jordan Curve Theorem

Overview

Simple connectivity underlies many basic results of complex analysis. Recall (Chapter 3) that a region Ω is *simply connected* if any two curves in Ω with the same end-points are homotopic; from this we deduced that

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every f holomorphic in Ω. In this appendix we prove:

(I) Ω is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all f holomorphic in Ω.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) Ω is simply connected if and only if no curve in Ω winds around a point of Ω^c.

In the second part we answer: given a simple closed curve γ, what is the region "enclosed by γ"? We prove the Jordan curve theorem for piecewise-smooth curves: γ^c has exactly two components, the bounded one is simply connected and both have boundary γ. This yields a general form of Cauchy's theorem extending the version for toy contours of Chapter 2. As in Chapter 1, "curve" means "piecewise-smooth curve" unless stated otherwise.

1 Equivalent formulations of simple connectivity

Motivation. Cauchy's theorem holds in a region when every closed curve can be shrunk to a point, but in practice one wants other ways to check this: by winding numbers, or by the complement being connected. This section proves that these conditions all agree.

Theorem 1.1

A region Ω is holomorphically simply connected if and only if Ω is simply connected.

Proof. If Ω is simply connected, it is holomorphically simply connected by Corollary 5.3 of Chapter 3. Conversely, let Ω be holomorphically simply connected. If $\Omega = \mathbb{C}$ it is simply connected. Otherwise the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim: the only use of simple connectivity there is to construct a holomorphic logarithm (hence square root) of a non-vanishing holomorphic function g, namely $\log g(z) = c + \int_{z_0}^{z} g'/g$, which is well defined precisely because (1) holds. Hence Ω is conformally equivalent to $\mathbb{D}$. Since $\mathbb{D}$ is simply connected and a conformal map carries homotopies to homotopies, Ω is simply connected. ∎

Theorem 1.2

If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if Ω^c is connected.

Boundedness is needed: an infinite strip is simply connected but its complement has two components. If complements are taken in the Riemann sphere, the conclusion holds for all regions.

Proof that Ω^c connected $\Rightarrow$ Ω simply connected. By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and f holomorphic in Ω. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

K_δ is compact (Ω is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small δ since η is compact. We show that K_δ^c is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with O_1, O_2 disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

Step F_1 is closed. Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ (Ω^c is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large n as O_2 is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise F_2 is closed.

Step $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. Pick $w \in O_1$. Since $w \notin K_\delta$, there is $z \in \Omega^c$ with $|w - z| < \delta$. Every point of the segment $[w, z]$ is within distance $< \delta$ of $z \in \Omega^c$, hence lies in K_δ^c. So the connected segment lies in $O_1 \cup O_2$ and meets both ($w \in O_1$, $z \in O_2$), a contradiction. Likewise $F_2 \neq \emptyset$.

Thus Ω^c is disconnected, contrary to hypothesis; so K_δ^c is connected. By Runge's theorem (Theorem 5.7 of Chapter 2) there are polynomials $P_n \to f$ uniformly on K_δ, hence on η. Since $\int_\eta P_n = 0$ (polynomials have primitives), $\int_\eta f = \lim \int_\eta P_n = 0$. ∎

The converse is proved after Theorem 1.4, using winding numbers.

Winding numbers

If γ is closed and $z \notin \gamma$, each loop of γ around z changes $\frac{1}{2\pi}\arg(\zeta - z)$ by ± 1. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

Definition (Winding number)

The **winding number** (or **index**) of a closed curve γ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for e^{-ikt} the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

Lemma 1.3

Let γ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If z, w lie in the same connected component of γ^c, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If z lies in the unbounded component of γ^c, then $W_\gamma(z) = 0$.

Proof. (i) Let $\gamma : [0,1] \to \mathbb{C}$ parametrize the curve and $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$. G is continuous and, except at finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H(t) = (\gamma(t) - z)e^{-G(t)}$ is continuous with $H' = \gamma' e^{-G} - (\gamma - z)G'e^{-G} = 0$ except at finitely many points, so H is constant: $H(1) = H(0) = \gamma(0) - z$. As $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$, so $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) W_γ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on z, uniformly on γ), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since W_γ is constant on the unbounded component by (ii), it is 0 there. ∎

Theorem 1.4

A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices to show Ω^c is connected. Suppose not: $\Omega^c = F_1 \cup F_2$ with F_1, F_2 disjoint, closed, non-empty. Only one of them can be unbounded (Ω is bounded, so Ω^c contains the connected set $\{|z| > R\}$), so we may assume F_1 is bounded, hence compact. We construct a closed curve $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using the following lemma. ∎

Lemma 1.5

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) w is in the interior of Q_1;

(ii) the interiors of Q_j and Q_k are disjoint for $j \neq k$;

(iii) F_1 is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from F_2;

(v) the boundary of $\bigcup_j Q_j$ lies in Ω and is a finite union of disjoint simple closed polygonal curves.

Proof of Theorem 1.4, assuming Lemma 1.5. Orient each ∂Q_j positively. Since w is interior to Q_1 and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

A side shared by two squares of $\mathcal{Q}$ is traversed twice in opposite directions and cancels. The remaining sides, oriented with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_{j=1}^{M} W_{\gamma_j}(w) = 1$. Hence $W_{\gamma_{j_0}}(w) \neq 0$ for some j_0, while $\gamma_{j_0} \subset \Omega$ and $w \notin \Omega$: a contradiction. ∎

Proof of Lemma 1.5. F_1 is compact and F_2 closed, so $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ such that w is the center of a square R_1, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be the squares of $\mathcal{G}_0$ meeting F_1.

Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$. (i), (ii) are clear. Every point of $\bigcup R_j$ is within $\sqrt{2}\delta < d$ of F_1, giving (iv). If $z \in F_1$, every square of $\mathcal{G}_0$ containing z meets F_1 and so belongs to $\mathcal{R}$; these squares cover a neighborhood of z, giving (iii). A *boundary side* (a side of exactly one square of $\mathcal{R}$) is a side of a square not in $\mathcal{R}$, so it misses F_1; it also misses F_2 by (iv). Hence $\partial \bigcup R_j \subset \Omega$.

Step Bad vertices. The boundary of $\bigcup R_j$ is the union of the boundary sides; their end-points are boundary vertices. At a grid vertex P the four surrounding squares give 0, 2 or 4 boundary sides at P, and 4 occurs exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not. Call such a P *bad*.

Step Refinement. Let $\mathcal{G}$ be the grid obtained by cutting each square of $\mathcal{G}_0$ into 9 equal subsquares, and take all $9m$ subsquares of squares of $\mathcal{R}$, with Q_1 the middle subsquare of R_1 (so w is its center). At each bad vertex P add the two subsquares of $\mathcal{G}$ having P as a corner and lying in the two missing squares of $\mathcal{G}_0$. Call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. Now all four small squares at P are in $\mathcal{Q}$, so P is interior. An added subsquare s at P has three other corners. At the two lying on sides of $\mathcal{G}_0$, three of the four surrounding small squares are in $\mathcal{Q}$. At the one inside a big square, only s is. So none of them is bad. Added squares at different bad vertices sit at different corners of a big square, so

they do not touch. Hence $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist: the added squares lie in squares of $\mathcal{G}_0$ adjacent to $\mathcal{R}$, within $3\delta < d$ of F_1. The new boundary sides lie in squares of $\mathcal{G}_0$ not in $\mathcal{R}$, so they miss F_1, and also F_2. So $\partial \bigcup Q_j \subset \Omega$.

Step (v). Every boundary vertex of $\bigcup Q_j$ now meets exactly two boundary sides. With the orientation leaving $\bigcup Q_j$ on the left, one of them enters the vertex and the other leaves it (check the cases of one, two adjacent, or three surrounding squares). Starting from a boundary side $[a_1, a_2]$ we get a unique continuation $[a_2, a_3], [a_3, a_4], \ldots$. There are finitely many sides, so some vertex repeats; let a_m be the first repetition, $a_m = a_k$ with $k < m$. If $k > 1$, then a_k meets the three distinct sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, which is impossible. So $k = 1$ and $a_1 \cdots a_m$ is a simple closed polygon. Removing its sides and repeating exhausts the boundary sides, giving finitely many disjoint simple closed polygons (disjoint because each vertex lies on only one of them). ■

Proof of Theorem 1.2, converse. Let Ω be bounded and simply connected. If Ω^c were disconnected, the proof of Theorem 1.4 produces a closed curve $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$. This contradicts the first part of Theorem 1.4, since Ω is simply connected. ■

2 The Jordan curve theorem

Motivation. Contour integration constantly refers to "the interior of a curve", and the Jordan curve theorem guarantees that such an interior exists for every simple closed curve. We prove it here and deduce a general form of Cauchy's theorem.

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves γ_η below).

Theorem 2.1 (Jordan arc theorem)

Let γ be a simple (non-closed) piecewise-smooth curve in the plane. Then γ^c is an open connected set whose boundary is precisely γ.

Theorem 2.2 (Jordan curve theorem)

Let γ be a simple closed piecewise-smooth curve in the plane. Then γ^c consists of two disjoint connected open sets. Exactly one of them is bounded and simply connected; it is the **interior** Ω_γ of γ. The other is unbounded; it is the **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of γ,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

Remark

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

Theorem 2.3

If f is holomorphic in the interior Ω_γ of a simple closed curve γ, then $\int_\eta f(\zeta)\, d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

Proof [Supplement]. Ω_γ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3 of Chapter 3. ■

Idea of the proof of Theorem 2.1. γ^c is open, so it suffices to show it is pathwise connected. Join $z, w \notin \gamma$ by a curve. If it meets γ, first go from z to a point z' and from w to a point w' near γ without meeting γ. Then join z' to w' by travelling "parallel" to γ, going around an end-point if necessary. If γ parametrizes a smooth piece, $\gamma'(t) \neq 0$ is tangent to it, $i\gamma'(t)$ is normal, and $\gamma(t) + i\eta\gamma'(t)$ is a "parallel" curve.

In the next three lemmas and two propositions, γ_0 is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'(t)| = 1$ (every smooth curve has one). An *interior point* of γ_0 is a point $\gamma(t)$ with $0 < t < L$ (not to be confused with the interior of Theorem 2.2).

Lemma 2.4

For $\eta \in \mathbb{R}$ let γ_η be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

Proof.

Step Local version. For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of γ' there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \ge |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |t - s + i\eta| - \frac{|t - s|}{2} \ge \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \neq \gamma(s)$ when $|t - s| < \delta$ and $\eta \neq 0$.

Step Global version. Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \le \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \ge \delta/2\}$ (intersected with $[0, L]$). The I_k cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \neq 0). \tag{3}$$

Since γ_0 is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |\gamma(t) - \gamma(s)| - |\eta| \ge d_k - |\eta| \ge d_k/2 \quad \text{if } |\eta| \le d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ∎

Lemma 2.5

Suppose $z \notin \gamma_0$ and $z_0 \in \gamma_0$ is a point closest to z with $z_0 = \gamma(t_0)$, $0 < t_0 < L$ (this holds if z is closer to an interior point of γ_0 than to either end-point). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$ for some real $\eta \neq 0$, i.e. $z \in \gamma_\eta$.

Proof. Near t_0, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \le |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\,[(z - \gamma(t_0))\overline{\gamma'(t_0)}] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \neq 0$ because $z \notin \gamma_0$. ∎

If $z \in \gamma_\eta$ and $w \in \gamma_\epsilon$ ($\eta, \epsilon \neq 0$) are obtained as in Lemma 2.5, we say z, w are on the **same side** of γ_0 if η, ϵ have the same sign, and on **opposite sides** otherwise. We do not define "the two sides of

γ_0"; the notion refers to the chosen representations, and we do not claim the two cases are exclusive. Points on the same side can be joined almost directly by a curve parallel to γ_0. Points on opposite sides require going around an end-point.

Proposition 2.6

Let A, B be the end-points of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A, B\}$ (or a subset thereof). If $z, w \notin \gamma_0$ lie on the same side of γ_0, and each is closer to an interior point of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $K \cup \gamma_0$.

Proof. By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to z, w,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with η_0, ϵ_0 of the same sign, say positive, and say $t_0 \le s_0$.

Step Normal segments. The half-open segment from z to z_0 lies in $(K \cup \gamma_0)^c$. If p is on it with $p \neq z_0$ and $p \in K \cup \gamma_0$, then $|z - p| < |z - z_0| = d(z, K \cup \gamma_0)$, which is absurd. So for all small $\eta > 0$, z can be joined to $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and likewise w to $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

Step Parallel curve. Choose $\eta < \eta_1$ (Lemma 2.4) and η smaller than the distance from K to $\{\gamma(t) : t_0 \le t \le s_0\}$. This distance is positive since this arc avoids A, B. Then $\{\gamma_\eta(t) : t_0 \le t \le s_0\}$ joins z_η to w_η. It misses γ_0 by Lemma 2.4, and it misses K since $|\gamma_\eta(t) - \gamma(t)| = \eta$. ∎

Lemma 2.7

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \le \theta \le \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from γ_0.

Proof. As in Lemma 2.4,

$$\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)]\, du + (L - t + \eta e^{i\theta})\gamma'(L).$$

Choose δ with $|\gamma'(u) - \gamma'(L)| < 1/2$ for $|u - L| < \delta$. Since $\cos\theta \ge 0$, $|L - t + \eta e^{i\theta}|^2 \ge (L-t)^2 + \eta^2 \ge \frac{1}{2}(L - t + \eta)^2$. So for $L - \delta < t \le L$,

$$|\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t)| \ge \frac{L - t + \eta}{\sqrt{2}} - \frac{L - t}{2} \ge \frac{\eta}{\sqrt{2}} > 0.$$

Thus $\gamma(t) \notin N$ for $L - \delta < t \le L$. Finally, take η_2 less than the distance from $\gamma(L)$ to $\{\gamma(t) : 0 \le t \le L - \delta\}$, which is positive since γ_0 is simple. ∎

Proposition 2.8

Let A be an end-point of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A\}$. If $z, w \notin \gamma_0$ are closer to interior points of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $\gamma_0 \cup K$.

Proof. If z, w are on the same side, apply Proposition 2.6. So let them be on opposite sides, and (reversing the parametrization if needed) $A = \gamma(0)$, so the other end-point $B = \gamma(L) \notin K$. Write

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) - i\epsilon_0\gamma'(s_0), \qquad \eta_0, \epsilon_0 > 0,$$

and let $\tau = \min(t_0, s_0) > 0$. Choose $\eta > 0$ smaller than η_1, η_2 (Lemmas 2.4, 2.7), and the distance from K to $\{\gamma(t) : \tau \le t \le L\}$ (positive since this arc avoids A). As in Proposition 2.6, join z to $\gamma_\eta(t_0)$ by a normal segment. Then follow γ_η from t_0 to L to reach $\gamma(L) + i\eta\gamma'(L)$. Then follow the half-circle $\gamma(L) + \eta e^{i\theta}\gamma'(L)$, θ from $\pi/2$ down to $-\pi/2$, inside N, to reach $\gamma(L) - i\eta\gamma'(L)$. Then follow $\gamma_{-\eta}$ from L back to s_0, and the normal segment to w. The parallel pieces miss γ_0 by Lemma 2.4; the half-circle misses γ_0 by Lemma 2.7. All pieces lie within η of $\{\gamma(t) : \tau \le t \le L\}$, so they miss K. ■

Proof of Theorem 2.1.

Step Boundary. $O = \gamma^c$ is open and $\partial O \subset \gamma$. Every point of a smooth piece of γ is a limit of points of O: by Lemma 2.4 (applied to a small sub-arc, and noting the rest of γ is at positive distance), $\gamma(t) + i\eta\gamma'(t) \notin \gamma$ for small $\eta \neq 0$. Since ∂O is closed and the smooth points are dense in γ, $\partial O = \gamma$.

Step Connectedness: induction on the number of smooth pieces. First let γ be smooth, with end-points A, B, and let $Z, W \notin \gamma$. Join Z to W by a polygonal path Λ avoiding A and B. If Λ misses γ we are done. Otherwise let p and q be the first and last points of Λ on γ; they are interior points of γ. Points z of Λ just before p satisfy $d(z, \gamma) \le |z - p| \to 0$, while $|z - A|, |z - B|$ stay bounded below. So we may take such a $z \notin \gamma$ closer to an interior point of γ than to A, B, joined to Z along Λ without meeting γ. Similarly take w just after q. Proposition 2.8 with $K = \emptyset$ joins z to w in γ^c.

Now suppose the result holds for curves of $n-1$ smooth pieces, and let $\gamma = K \cup \gamma_0$ with K the union of $n-1$ consecutive pieces and γ_0 smooth. Then K is compact and $K \cap \gamma_0$ is a single end-point A of γ_0. Given $Z, W \notin \gamma$, the induction hypothesis gives a curve from Z to W missing K. Since K^c is open and connected, we may take it polygonal and avoiding the other end-point of γ_0. If it meets γ_0, it does so at interior points. As above, take z (resp. w) just before the first (resp. after the last) meeting point; it is closer to an interior point of γ_0 than to K or to the end-points, since its distance to K is bounded below. Proposition 2.8 joins z to w in $(\gamma_0 \cup K)^c = \gamma^c$. ■

Proof of Theorem 2.2.

Step At most two components. Fix W outside a large disc containing γ, and let $\mathcal{U}$ be the set of points joined to W by a continuous curve in γ^c. $\mathcal{U}$ is open and connected, and contains the exterior of the disc. Let $\Omega = \gamma^c - \mathcal{U}$; we show Ω is connected. Remove a smooth piece γ_0 (with end-points A, B) from γ, and let K be the remaining arc, so $K \cap \gamma_0 = \{A, B\}$. For $Z \in \Omega$, Theorem 2.1 applied to K gives a curve Λ_Z from Z to W missing K. Since $Z \notin \mathcal{U}$, Λ_Z meets γ_0, necessarily at interior points. As in the proof of Theorem 2.1, choose $z, w \in \Lambda_Z$ closer to interior points of γ_0 than to K or to A, B, such that the pieces of Λ_Z from Z to z and from w to W lie in γ^c. Then z, w are on opposite sides of γ_0. Otherwise Proposition 2.6 would join z to w in γ^c, hence Z to W, contradicting $Z \notin \mathcal{U}$. If $Z_1 \in \Omega$ is another point, with corresponding z_1, w_1 on opposite sides, then z and z_1 are on the same side. Otherwise z_1 is on the same side as w, and Proposition 2.6 joins Z_1 to W in γ^c, a contradiction. So Proposition 2.6 joins z to z_1 in γ^c, and Z, Z_1 lie in the same component. Thus γ^c has at most two components, $\mathcal{U}$ and (if non-empty) Ω.

Step Exactly two components. By Lemma 1.3 it suffices to find points with different winding numbers. Parametrize γ and fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$. For $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small ϵ these avoid γ, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \le \epsilon\}$ lie in γ^c (Lemma 2.4 for the piece, plus the positive distance from $\gamma(t_0)$ to the rest of γ). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t-t_0) + o(|t-t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v\gamma'(t)}{a(t)^2+\epsilon^2 v^2} = \frac{2i\epsilon}{(t-t_0)^2+\epsilon^2}\,(1+\rho(t,\epsilon)),$$

where, given $\sigma > 0$, there is $\delta > 0$ with $|\rho(t,\epsilon)| < \sigma$ for $|t-t_0| \le \delta$, uniformly in ϵ. (Indeed $a^2+\epsilon^2v^2 = v^2[(t-t_0)^2+\epsilon^2] + o((t-t_0)^2)$ and $o((t-t_0)^2)/[(t-t_0)^2+\epsilon^2] \to 0$.) Split the integral defining $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ into $|t-t_0| \ge \delta$ and $|t-t_0| < \delta$. On the first part $\gamma(t)$ stays at positive distance from $\gamma(t_0)$, while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part tends to 0 as $\epsilon \to 0$. On the second,

$$\frac{1}{2\pi i}\int_{|t-t_0|<\delta} \frac{2i\epsilon\,dt}{(t-t_0)^2+\epsilon^2} = \frac{1}{\pi}\int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2+1} = \frac{2}{\pi}\arctan(\delta/\epsilon) \to 1,$$

and the ρ-term is at most $\frac{\sigma}{\pi}\int_{\mathbb{R}} \frac{ds}{s^2+1} = \sigma$ in absolute value. Letting $\epsilon \to 0$ gives $|\Delta - 1| \le \sigma$ for every $\sigma > 0$, so $\Delta = 1$.

Hence z_ϵ and w_ϵ lie in different components, so γ^c has exactly the two components $\mathcal{U}$ and Ω. $\mathcal{U}$ is unbounded with $W_\gamma = 0$ there (Lemma 1.3(iii)), and Ω is bounded. Since $\Delta = 1$, $W_\gamma = \pm 1$ on Ω, and after possibly reversing orientation $W_\gamma = 1$ there. Every smooth point of γ is a limit of the z_ϵ and of the w_ϵ, hence of points of both components. Smooth points are dense in γ, and the boundary of each component lies in γ. So $\gamma = \partial\Omega = \partial\mathcal{U}$. Write $\Omega_\gamma = \Omega$ and $\mathcal{U}_\gamma = \mathcal{U}$.

Step Ω_γ is simply connected. By Theorem 1.2 it suffices that $\Omega_\gamma^c = \gamma \cup \mathcal{U}_\gamma$ is connected. Suppose $\Omega_\gamma^c = F_1 \cup F_2$ with F_1, F_2 closed, disjoint, non-empty, and let $O_i = \mathcal{U}_\gamma \cap F_i$. The sets O_1, O_2 are disjoint and $\mathcal{U}_\gamma = O_1 \cup O_2$.

O_1 is open. Let $z \in O_1$; small balls B at z lie in $\mathcal{U}_\gamma \subset F_1 \cup F_2$. If every such ball met F_2, then $z \in F_2$ (F_2 closed), contradicting $F_1 \cap F_2 = \emptyset$. So some ball lies in F_1, hence in O_1. Likewise O_2 is open.

$O_1 \neq \emptyset$. Otherwise $F_1 \subset \gamma$ and $\mathcal{U}_\gamma \subset F_2$. Take $z \in F_1 \subset \gamma$. Since $\gamma = \partial\mathcal{U}_\gamma$, every ball at z meets $\mathcal{U}_\gamma \subset F_2$, so $z \in F_2$, a contradiction. Likewise $O_2 \neq \emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ■

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9

If f is holomorphic in an open set containing a simple closed piecewise-smooth curve γ and its interior, then $\int_\gamma f = 0$.

Let $\mathcal{O} \supset \gamma \cup \Omega_\gamma$ be open with f holomorphic in $\mathcal{O}$. The idea is to build a closed curve $\Gamma \subset \Omega_\gamma$ so close to γ that $\int_\gamma f = \int_\Gamma f$. The right side vanishes by Theorem 2.3, since f is holomorphic in the simply connected set Ω_γ. Near smooth parts of γ, Γ is essentially a parallel curve γ_η (Lemma 2.4); near the junctions of smooth parts, Γ follows arcs of circles.

Lemma 2.10

Let $\gamma : [0,1] \to \mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon > 0$ the circle C_ϵ of center $\gamma(0)$ and radius ϵ meets γ in exactly one point.

Proof. Assume $\gamma(0) = 0$. Since $\gamma(1) \neq 0$, the intermediate value theorem applied to $|\gamma(t)|$ gives at least one intersection point for $\epsilon < |\gamma(1)|$. Suppose the claim fails: there are $\epsilon_j \to 0$ such that $|\gamma(t)| = \epsilon_j$ has two distinct solutions. Since $|\gamma(t)| \ge c(\tau) > 0$ for $t \ge \tau$ (simplicity), both solutions tend to 0. By the mean value theorem applied to $h(t) = |\gamma(t)|^2$, there are $t_j \to 0$, $t_j > 0$, with

$h'(t_j) = 2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}] = 0$. But $\gamma(t) = \gamma'(0)t + t\varphi(t)$ and $\gamma'(t) = \gamma'(0) + \psi(t)$ with $\varphi, \psi \to 0$ as $t \to 0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}] = |\gamma'(0)|^2 t + o(t),$$

which is $\neq 0$ for small $t > 0$ since $\gamma'(0) \neq 0$. This is a contradiction. ■

Proof of Theorem 2.9. Choose ϵ so small that the open set U of points at distance $< \epsilon$ from γ is contained in $\mathcal{O}$. Let $P_1, \ldots, P_n$ be the consecutive junction points of the smooth parts of γ. Choose $\delta < \epsilon/10$ so small that each circle C_j centered at P_j of radius δ meets γ in exactly two points. This is possible by Lemma 2.10, applied to the two smooth parts meeting at P_j; the rest of γ is at positive distance from P_j. The two points determine two arcs of C_j, exactly one of which, $\mathcal{C}_j$, has its interior in Ω_γ. Indeed, for a smooth part with end-point P_j and small η, the curves γ_η and $\gamma_{-\eta}$ lie on opposite sides of γ, hence in different components of γ^c (proof of Theorem 2.2), and both meet C_j. The disc D_j of center P_j and radius 2δ lies in U, hence in $\mathcal{O}$.

Let B_j be the end-point of $\mathcal{C}_j$ on the smooth part γ_j from P_j to P_{j+1}, and A_{j+1} the end-point of $\mathcal{C}_{j+1}$ on γ_j. Take a chain of discs $\mathcal{D} = \{D_0, \ldots, D_K\}$ in U covering γ, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs D_j among them. By Lemma 2.4 there is a continuous curve Γ_j in Ω_γ and in the union of the discs, running parallel to γ_j, from a point near B_j on $\mathcal{C}_j$ to a point near A_{j+1} on $\mathcal{C}_{j+1}$. Since γ is only C^1, Γ_j need not be smooth; replacing it by a nearby polygonal line we may assume it is. Joining Γ_j to Γ_{j+1} along a piece of $\mathcal{C}_{j+1}$, and so on, gives a closed piecewise-smooth curve $\Gamma \subset \Omega_\gamma$.

f has a primitive on each disc of $\mathcal{D}$ (Theorem 2.1 of Chapter 2). Arguing as in the proof of Theorem 5.1 of Chapter 3, and comparing γ and Γ disc by disc, gives $\int_\gamma f = \int_\Gamma f$: on consecutive discs the primitives differ by constants, and the endpoint contributions telescope. Since $\Gamma \subset \Omega_\gamma$ and Ω_γ is simply connected, $\int_\Gamma f = 0$ by Theorem 2.3. Hence $\int_\gamma f = 0$. ■